Toward Zero-shot Anomaly Detection

WACV 2025

TUCSON, ARIZONA • FEB 28 - MAR 4

2nd Workshop on Automated Spatial and Temporal Anomaly Detection (ASTAD)

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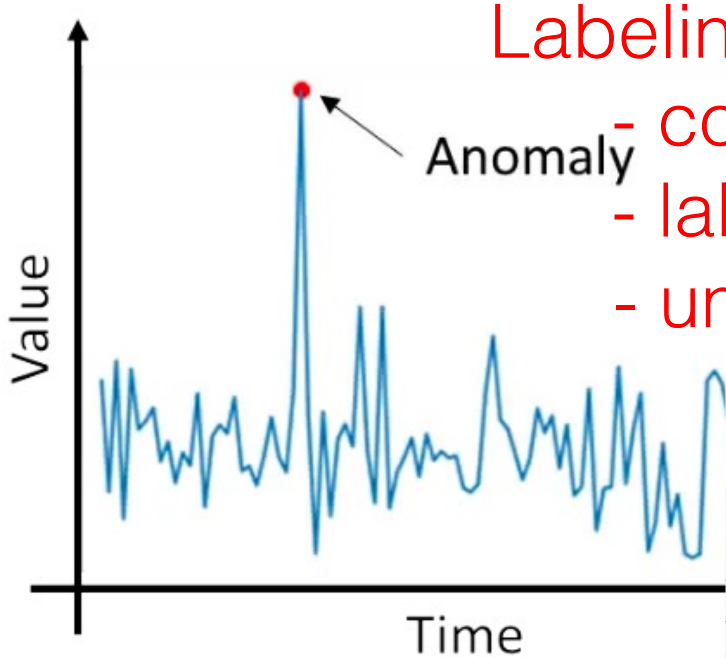
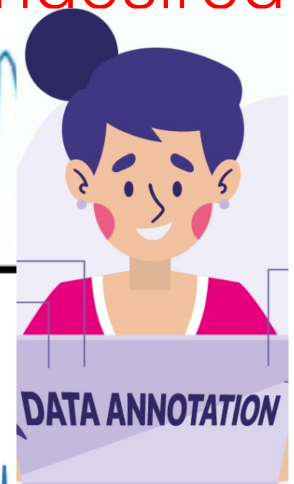
Anomaly Detection (AD)

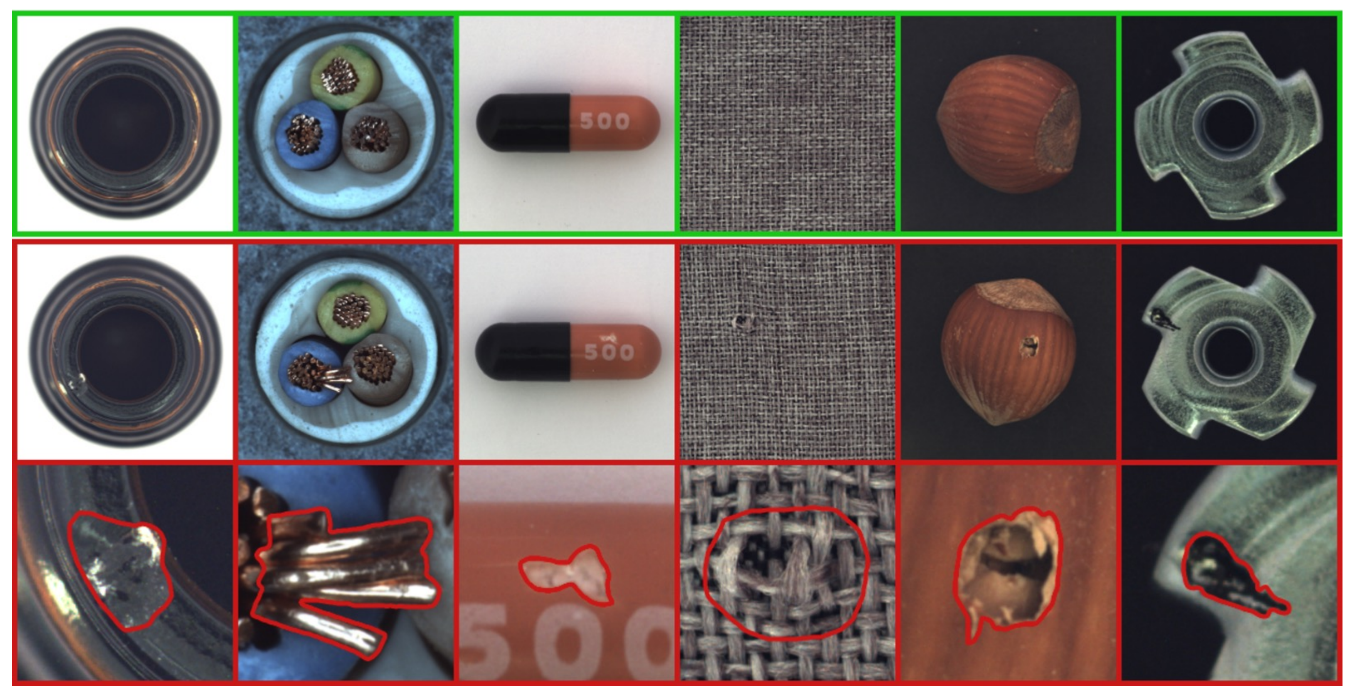
- **Anomaly detection (AD)** is to find **anomalies** from a set of data
  - **Unsupervised:** No information about actual anomalies



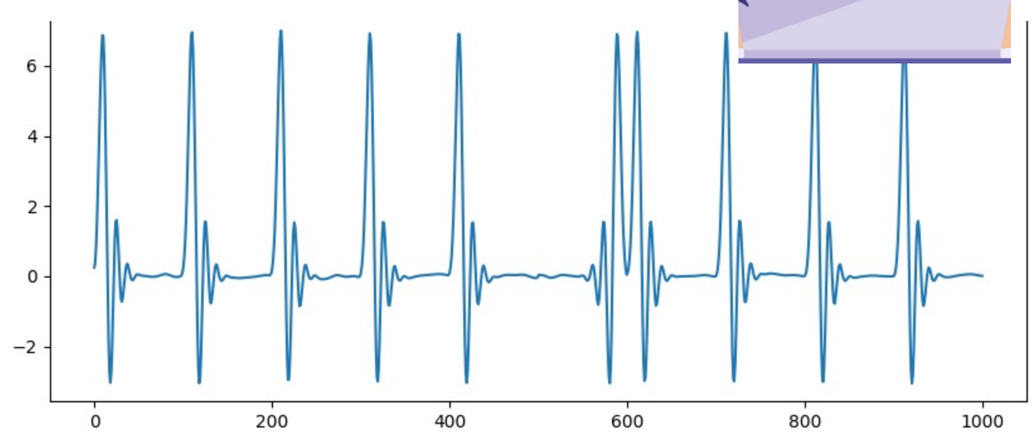
Source: Analytics Vidhya

Labeling is
- costly
- laborious
- undesired?



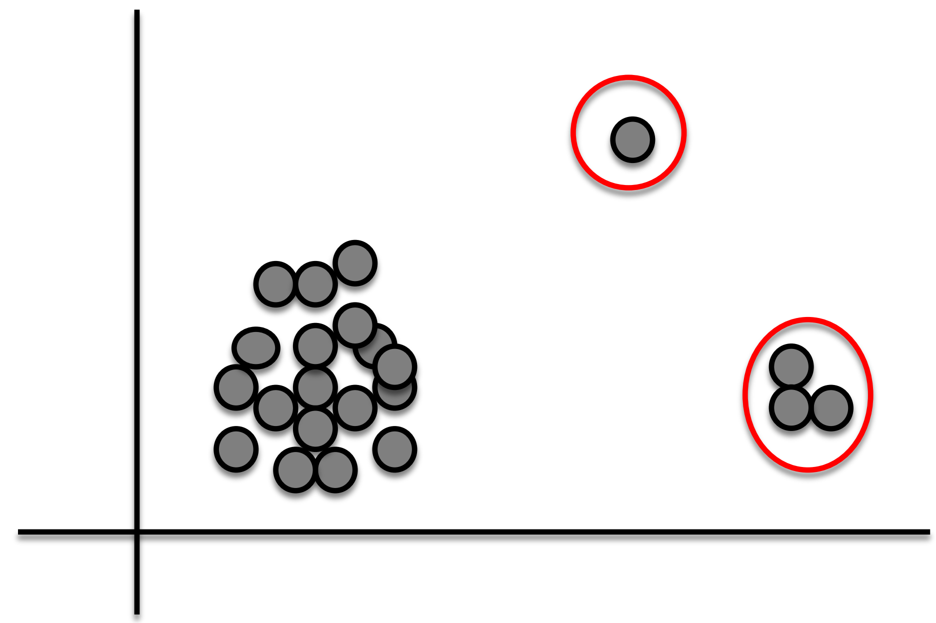


Source: https://www.mvtec.com/company/research/datasets/mvtec-ad



Outlier Detection Library

- **VAST number of detection methods**
  - ➢ Distance based (e.g. kNN)
  - ➢ Density based (e.g. LOF)
  - ➢ Depth based
  - ➢ Clustering based
  - ➢ Model/Stat. based
  - ➢ Angle based
  - ➢ Ensemble-based (e.g. iForest)
  - ➢ … most recently Deep Learning-based



Outlier Analysis. [Aggarwal] Springer, 2017

Deep Learning for Anomaly Detection: A Review. [Pang+] July 2020

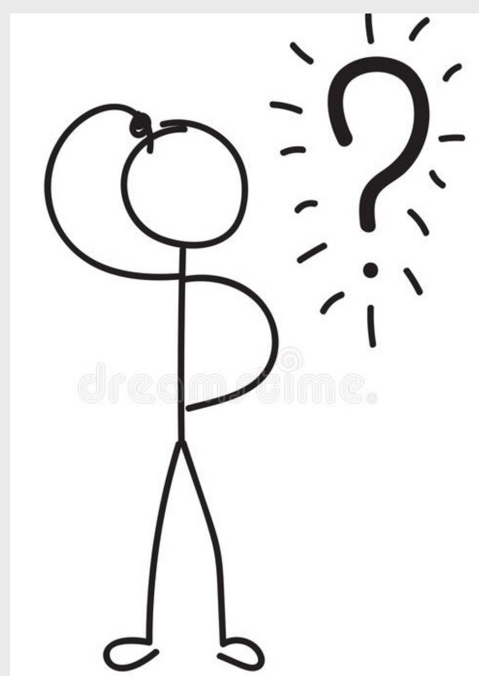
A Unifying Review of Deep and Shallow Anomaly Detection. [Ruff+] Sep. 2020

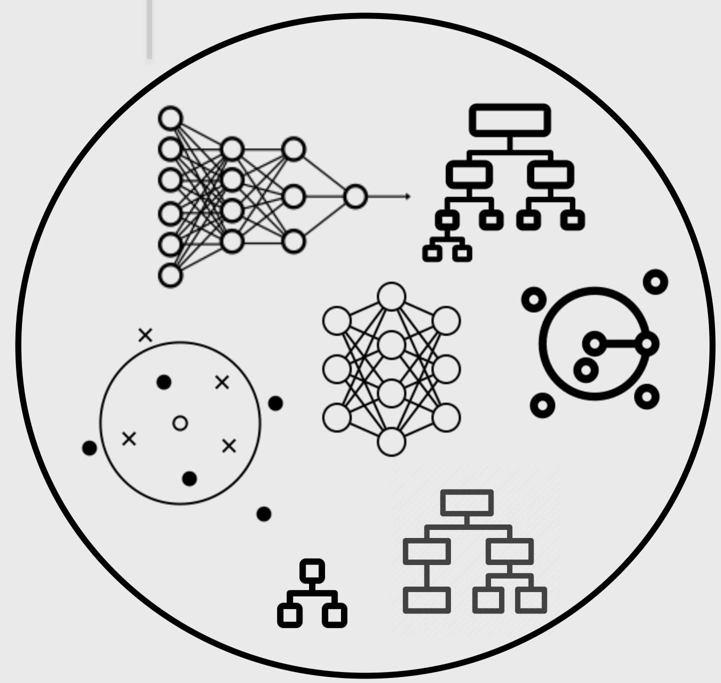
Unsupervised Model Selection

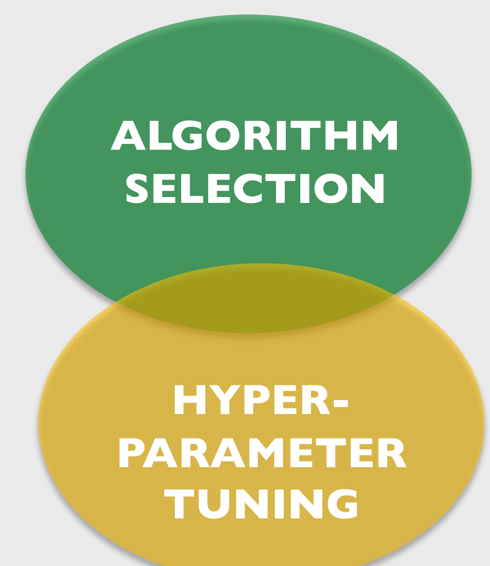
Given an **unsupervised**
anomaly detection task/dataset,

HOW TO DECIDE WHICH MODEL TO USE?

Accounting



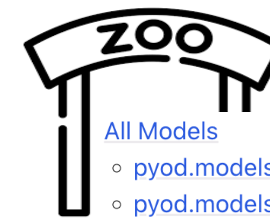




OD Library

- **Many classical (shallow) methods …**
  - **Algo Families** : Distance based, Density based, Depth based, Clustering based, Model/Stat. based, Angle based, Ensemble-based, …
  - … with **1-3** hyperparameters **(HPs)**

Algo + HP(s) → Model



All Models

- pyod.models.abod module
- pyod.models.alad module
- pyod.models.anogan module
- pyod.models.auto_encoder module
- pyod.models.auto_encoder_torch module
- pyod.models.cblof module
- pyod.models.cof module
- pyod.models.combination module
- pyod.models.cd module
- pyod.models.copod module
- pyod.models.deep_svdd module
- pyod.models.ecod module
- pyod.models.feature_bagging module
- pyod.models.gmm module
- pyod.models.hbos module
- pyod.models.iforest module
- pyod.models.inne module
- pyod.models.kde module
- pyod.models.knn module
- pyod.models.lmdd module
- pyod.models.loda module
- pyod.models.lof module
- pyod.models.loci module
- pyod.models.lunar module
- pyod.models.lscp module
- pyod.models.mad module
- pyod.models.mcd module
- pyod.models.mo_gaal module
- pyod.models.ocsvm module
- pyod.models.pca module
- pyod.models.rgraph module
- pyod.models.rod module
- pyod.models.sampling module
- pyod.models.sod module
- pyod.models.so_gaal module
- pyod.models.sos module
- pyod.models.suod module
- pyod.models.vae module
- pyod.models.xgbod module

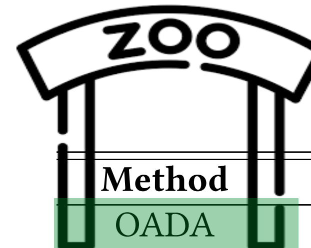
Src. https://pyod.readthedocs.io/en/latest/pyod.html

OD Library (cont.)

- … and now **DeepLearning (DL) based OD** is booming:
  - **Families** : One-class, AutoEncoder, GAN-based, Self-supervised, …
  - … **with >= 8 (!) HPs**, 3 groups:
    - i. Architecture: depth, width, …
    - ii. Regul.: weight decay, dropout, …
    - iii. Opt.: learning rate, #epochs, …

Algo + HP(s) → Model

Pang, G., Shen, C., Cao, L. and Hengel, A.V.D. Deep learning for anomaly detection: A review. ACM Computing Surveys (CSUR), (2021).



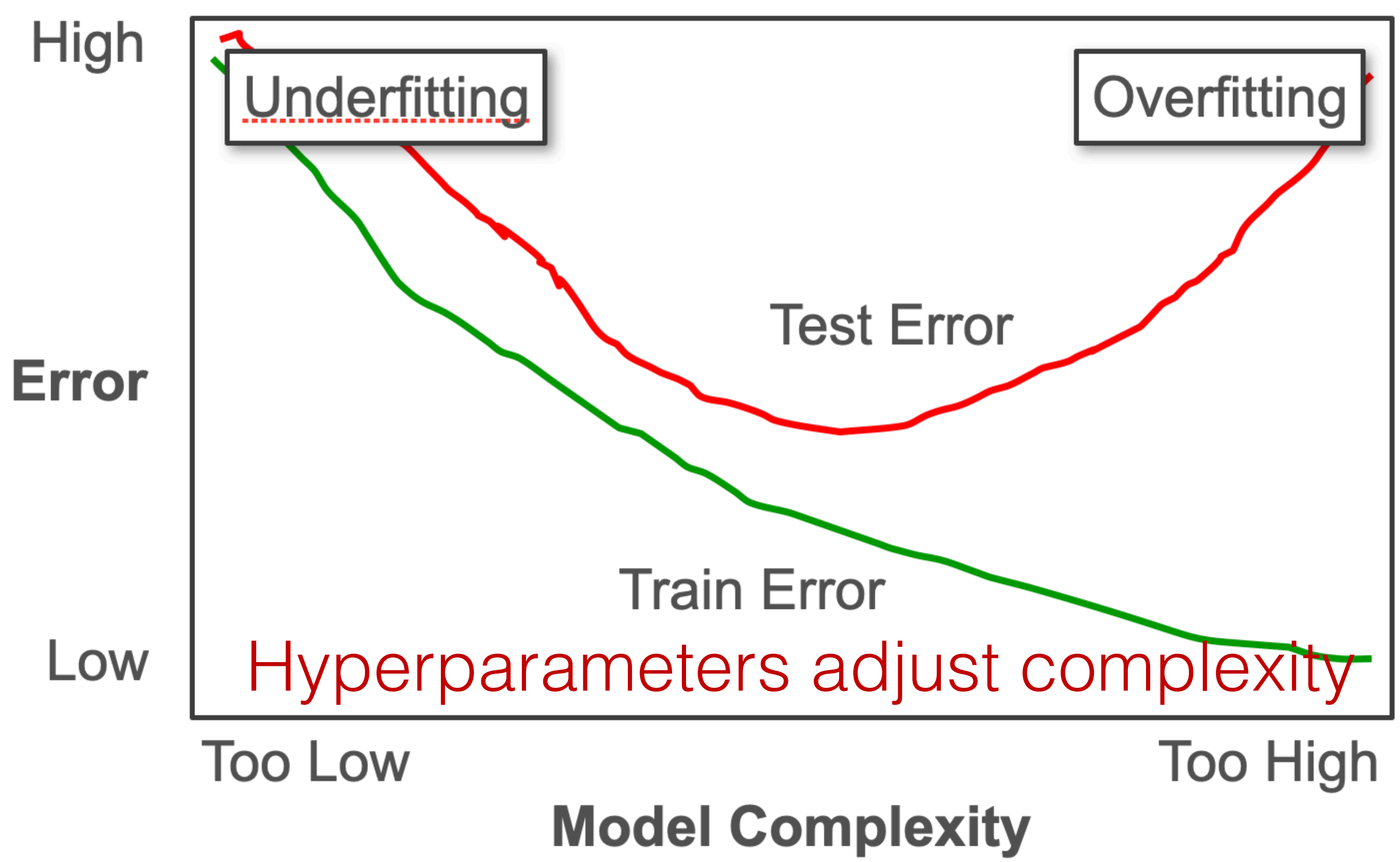
Method	Ref.	Sup.	Objective	DA
OADA	[66] (4)	Semi	Reconstruction	Yes
Replicator	[58] (5.1.1)	Unsup.	Reconstruction	No
RandNet	[29] (5.1.1)	Unsup.	Reconstruction	No
RDA	[176] (5.1.1)	Semi	Reconstruction	No
UODA	[93] (5.1.1)	Semi	Reconstruction	No
AnoGAN	[139] (5.1.2)	Semi	Generative	No
EBGAN	[171] (5.1.2)	Semi	Generative	No
FFP	[88] (5.1.3)	Semi	Predictive	Yes
LSA	[1] (5.1.3)	Semi	Predictive	No
GT	[49] (5.1.4)	Semi	Classification	Yes
E^3Outlier	[159] (5.1.4)	Semi	Classification	Yes
REPEN	[114] (5.2.1)	Unsup.	Distance	No
RDP	[157] (5.2.1)	Unsup.	Distance	No
AE-1SVM	[106] (5.2.2)	Unsup.	One-class	No
DeepOC	[162](5.2.2)	Semi	One-class	No
Deep SVDD	[133] (5.2.2)	Semi	One-class	No
Deep SAD	[134] (5.2.2)	Semi	One-class	No
DEC	[163] (5.2.3)	Unsup.	Clustering	No
DAGMM	[179] (5.2.3)	Unsup.	Clustering	No
SDOR	[118] (6.1)	Unsup.	Anomaly scores	No
PReNet	[115] (6.1)	Weak	Anomaly scores	Yes
MIL	[146] (6.1)	Weak	Anomaly scores	No
PUP	[109] (6.2)	Unsup.	Anomaly scores	No
DevNet	[116] (6.2)	Weak	Anomaly scores	No
APE	[30] (6.3)	Unsup.	Anomaly scores	No
AEHE	[46] (6.3)	Unsup.	Anomaly scores	No
ALOCC	[136] (6.4)	Semi	Anomaly scores	Yes
OCAN	[175] (6.4)	Semi	Anomaly scores	No
Fence GAN	[105] (6.4)	Semi	Anomaly scores	No
OCGAN	[121] (6.4)	Semi	Anomaly scores	No

Why Model Selection?

ML (Test) Error = Bias + Variance + Bayes Noise

Complexity of feature space & Model complexity

Discriminability of examples



Src. Arun Kumar, https://adalabucsd.github.io/papers/2021_DLDelusions_KDD_Slides.pdf

Roadmap

1. Unsupervised Learning
   Hyperparameter sensitivity in OD
2. Self-Supervised (SS) Learning
   toward Self-Tuning OD
3. Zero-shot OD: Model Selection Bygone?

Roadmap

1. Unsupervised Learning
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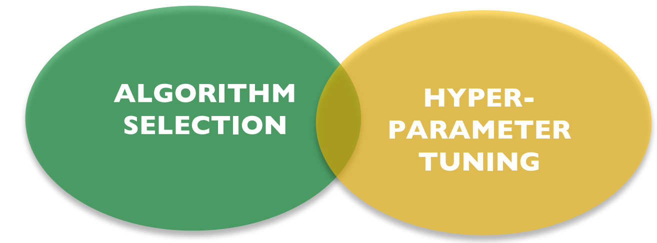
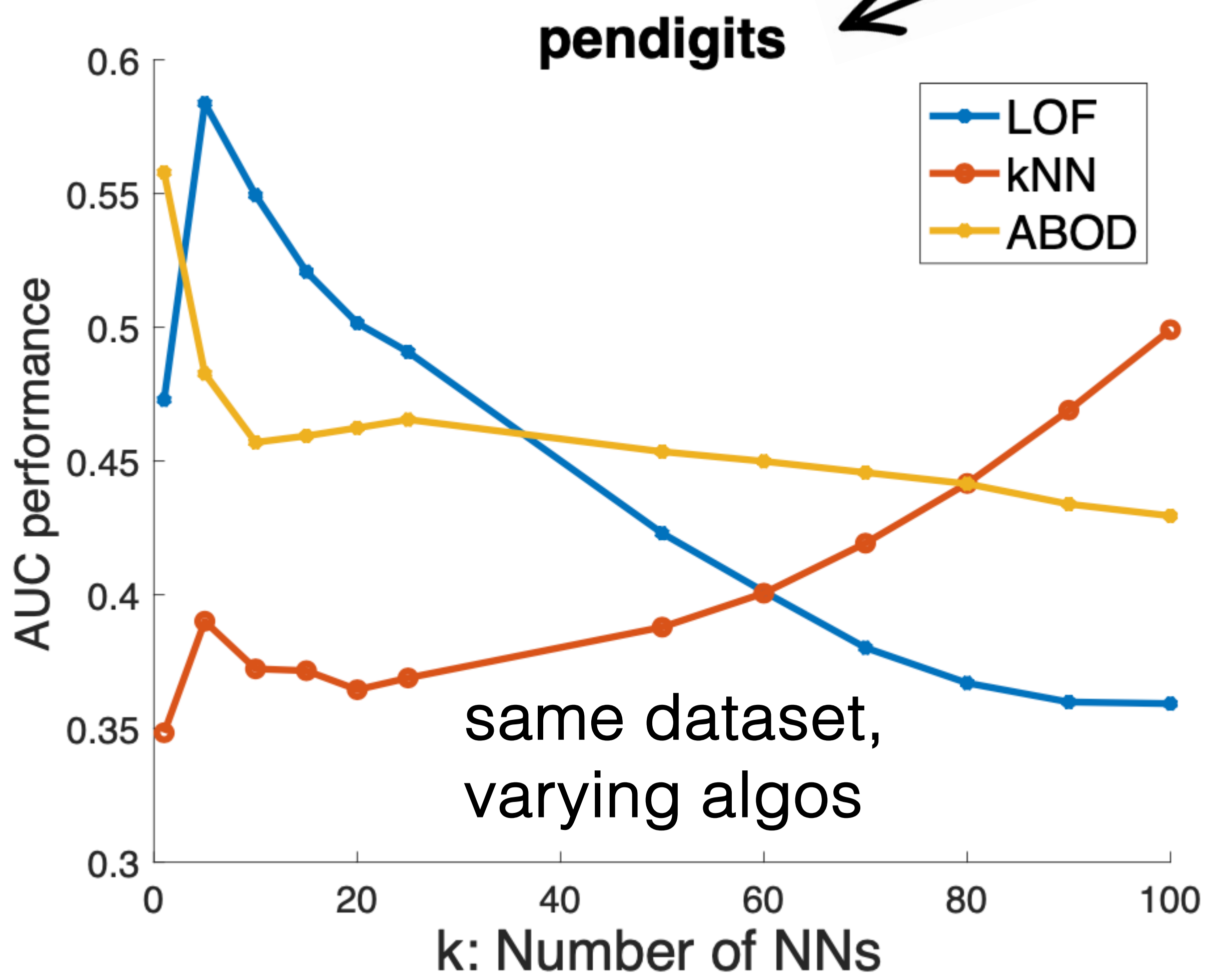
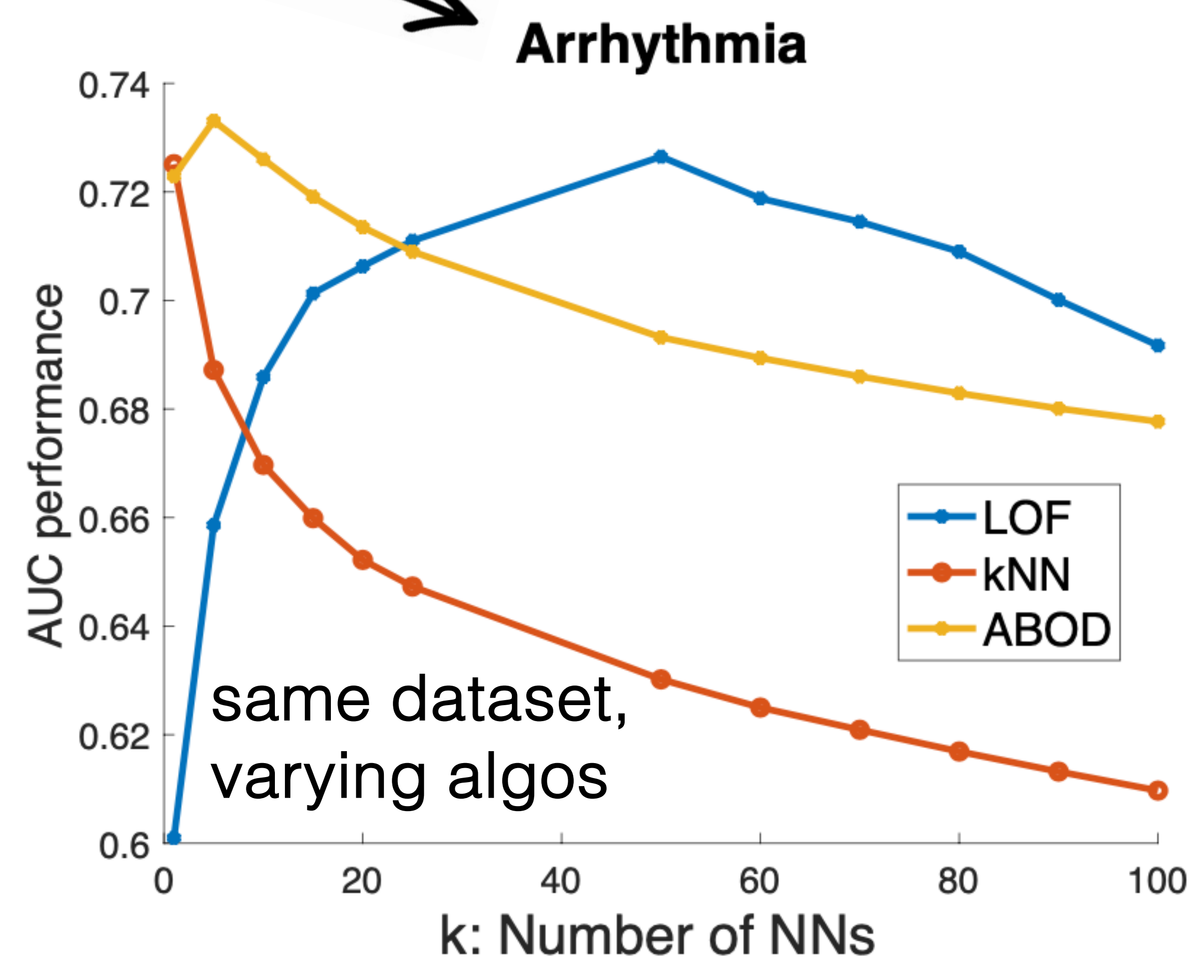
Why OD Model Selection?

- **OD algorithms are sensitive** to HPs.



same algo, varying datasets





Nearest-neighbors based methods: kNN, LOF, ABOD

MetaOD: Automating Outlier Detection via Meta-Learning.
Yue Zhao, Ryan Rossi, Leman Akoglu.
NeurIPS 2021

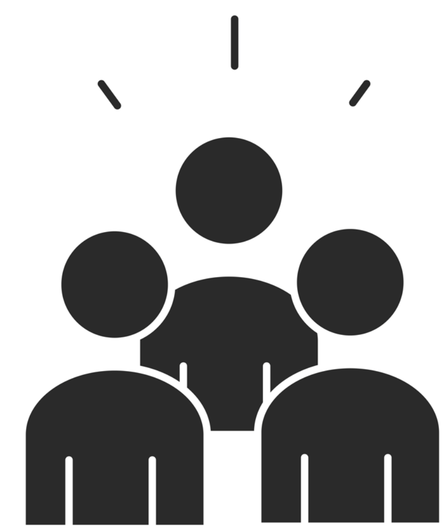
Who cares & Why?

- In the real world:

  Practitioner aims to
  carefully choose algo & HP(s)
  to do well on their (new) task.

- In academia:

  Research community aims for
  a fair comparison of methods
  & keep accurate record of progress
  on a research problem.



Is OD Model Selection Hard?

- in (supervised) ML: To measure generalization performance (test error):
  - Evaluate each model on hold-out (labeled) data
  - Pick the model with best validation-data performance

Is OD Model Selection Hard?

- in (supervised) ML: To measure generalization performance (test error):
  - Evaluate each model on hold-out (validation) data
  - Pick the model with best validation-data performance

! No labeled data to hold-out for validation

! No universally-accepted loss function,

- in fact, some detectors are **measure-based**
- **loss may mislead** – e.g. an auto-encoder overfit to outliers have better loss

- Large search space (for modern/deep models)

The State of MS in Classical OD community

- In the literature, several Eval / Benchmarking surveys on classical methods highlight the sensitivity issue to HPs :

  Markus Goldstein and Seiichi Uchida. A comparative evaluation of unsupervised anomaly detection algorithms for multivariate data. PloS One, 2016.

  Guilherme O. Campos, Arthur Zimek, Jörg Sander, Ricardo J. G. B. Campello, Barbora Micenková, Erich Schubert, Ira Assent, and Michael E. Houle. On the evaluation of unsupervised outlier detection: measures, datasets, and an empirical study. ACM DAMI , 2016.

  Charu C. Aggarwal and Saket Sathe. Theoretical foundations and algorithms for outlier ensembles. SIGKDD Explor., 2015.

- Others designed internal (proxy) evaluation measures:

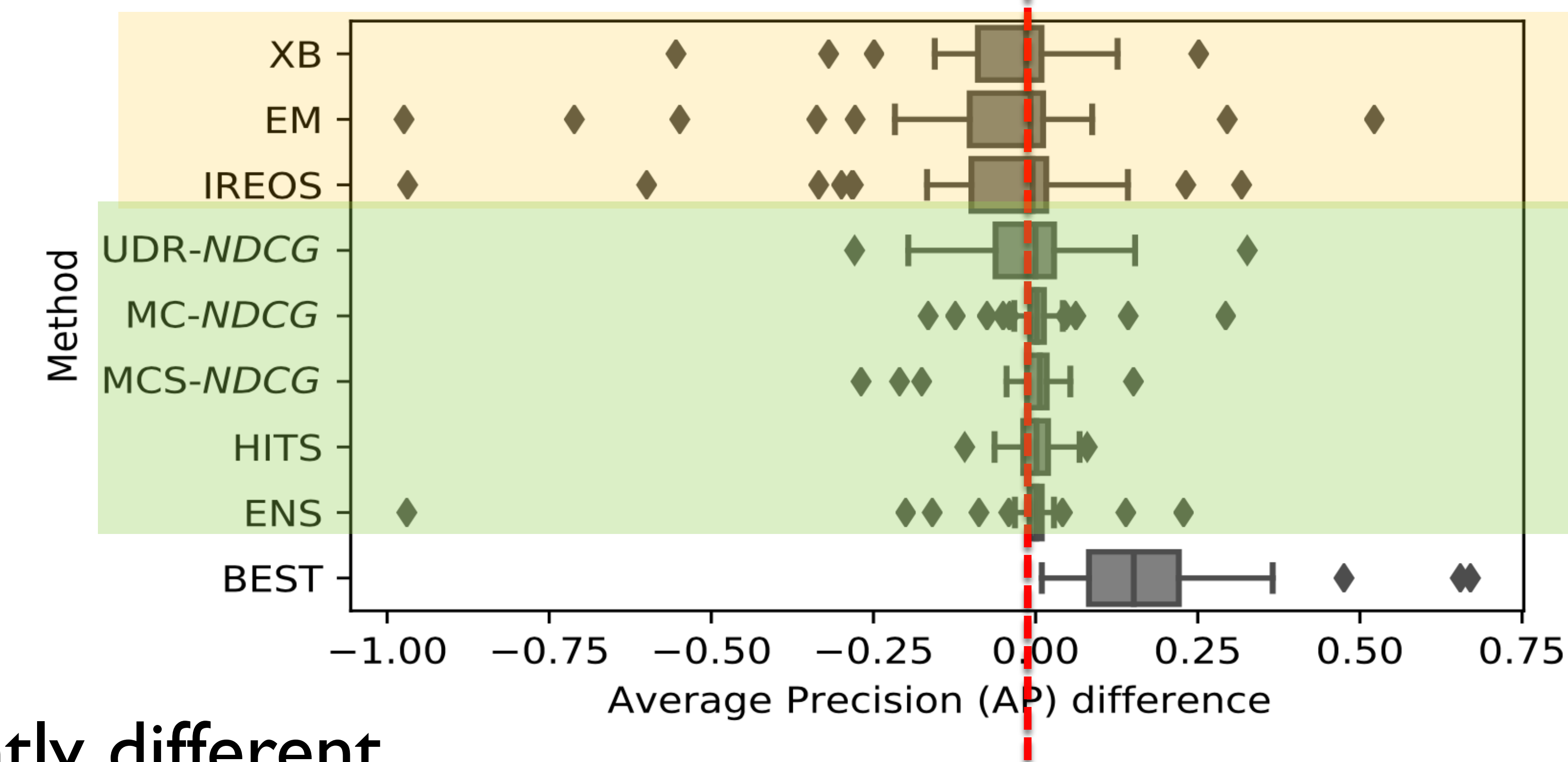
  Marques, H. O., Campello, R. J. G. B., Sander, J., and Zimek, A. Internal evaluation of unsupervised outlier detection. ACM TKDD, 2020.

  Goix, N. How to evaluate the quality of unsupervised anomaly detection algorithms? CoRR, abs/1607.01152, 2016.

  Nguyen, V., Nguyen, T., and Nguyen, U. An evaluation method for unsupervised anomaly detection algorithms. J. of CS and Cybernetics, 2017.

The State of MS in Classical OD community

- Importantly, our study (below) showed **none of them would be useful in practice**:



Not significantly different
from Random (stand-alone ones),
from iForest w/ default HPs (consensus-based ones)

A Large-scale Study on Unsupervised Outlier Model Selection: Do Internal Strategies Suffice? M. Ma, Y. Zhao, X. Zhang, L. Akoglu
ACM SigKDD Explorations Newsletter arXiv:2104.01422 (2021).

The State of MS
¯_(ツ)_/¯ Deep OD community

Table 6: Representative unsupervised deep OD models from 4 different families (for a broader coverage, see surveys [30, 34, 9]), annotated in terms of data used for training, test, and validation/model selection (if any). No existing work attempts (unsupervised) model selection; vast majority reports results for a *fixed* (how, unclear) "recommended" config. or tune *some* but not all HPs using labeled validation and even test (!) data. AE: autoencoder, SSL: self-supervised learning, Clean: inlier-only.

Method	**Year**	**Family**	**Train**	**Test**	**Validation (HP/Model Selection)**
RandNet [11]	2017	AE ensemble	Polluted	=Train	None, fixed – sensitivity analysis on some HPs
RDA [45]	2017	AE	Polluted	=Train	Best λ on Test, other HPs fixed
DAGMM [46]	2018	AE & density	Clean & Pol.d	Disjoint	None, fixed – sensitivity on reg. param.s $\{\lambda_1, \lambda_2\}$
DeepSVDD [33]	2018	One-Class	Clean	Disjoint	Best ν on Test, other HPs fixed
DROCC [18]	2020	One-Class	Clean	Disjoint	Validation data to tune some (not all) HPs
HRN [23]	2020	One-Class	Clean	Disjoint	10% of Test for tuning $\{\lambda, n\}$, other HPs fixed
(f-)AnoGAN [35, 36]	2017	GAN	Clean & Pol.d	Disjoint	None, fixed
EGBAD [43]	2018	(Bi)GAN	Clean	Disjoint	None, fixed
GANomaly [4]	2018	GAN	Clean	Disjoint	Best reg. weights $\{w_{adv}, w_{con}, w_{enc}\}$ on Test, others fixed
GOAD [5]	2020	SSL	Clean	Disjoint	None, fixed
NeuTraL [6]	2020	SSL	Clean	Disjoint	10% of Test for tuning transformation HPs, others fixed

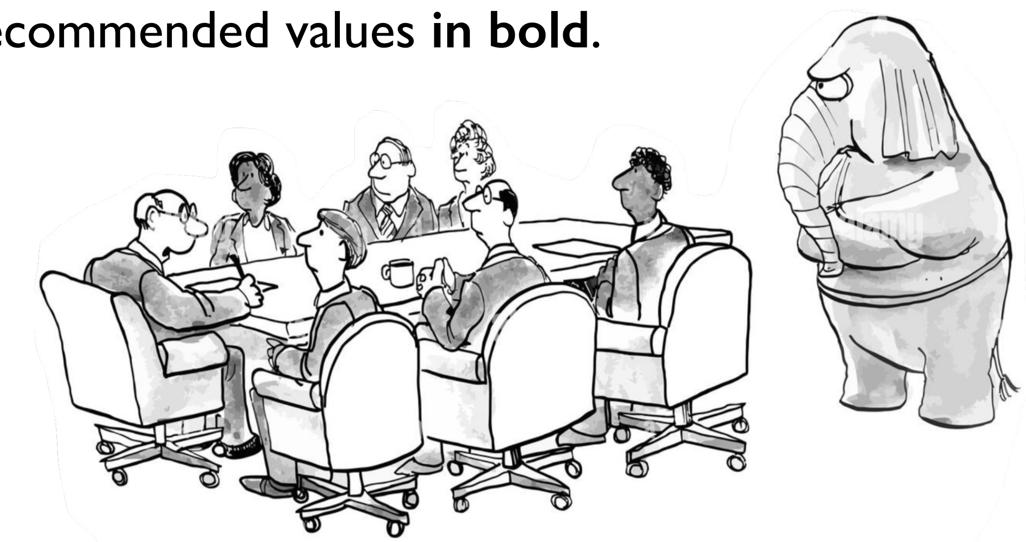
Hyperparameter Sensitivity in Deep Outlier Detection: Analysis and a Scalable Hyper-Ensemble Solution.
Xueying Ding, Lingxiao Zhao, Leman Akoglu.
NeurIPS 2022

Method	Hyperparameter	Grid	#values
VanillaAE	n_layers	[2, 3, 4]	3
	layer_decay	[1, 2, 4]	3
	LR	[1e-3, 1e-4, 1e-5]	3
	iter	[200, 500, 1000]	3
		Total =	81
RDA	λ	[5e-1, 5e-3, 5e-5]	3
	n_layers	[2, 3, 4]	3
	layer_decay	[1, 2, 4]	3
	LR	[1e-3, 1e-4]	2
	inner_iter	[20, 50]	2
	iter	[5, 20, 50]	3
		Total #models =	324
GANomaly	w_{adv}	**1**	1
	w_{con}	[25, **50**, 100]	3
	w_{enc}	[0.1, **1**]	2
	z_dim	[50, **100**, 200]	3
	LR	[5e-3, **2e-3**, 5e-4]	3
	iter	[10, **15**, 25]	3
		Total #models =	162

Method	Hyperparameter	Grid	#values
DeepSVDD	conv_dim	[**8**, 16, 32]	3
	fc_dim	[**16**, 32]	2
	Relu_slope	[**1e-1**, 1e-3]	2
	pretr_iter	[200, **350**, 400]	3
	pretr_LR	[1e-4, 1e-5]	2
	iter	[100, 200, **250**]	3
	LR	[1e-4, 1e-5]	2
	wght_dc	[1e-5, **1e-6**]	2
		Total #models =	864
RandNet	n_layers	[3, 5, **7**, 9]	4
	layer_decay	[0.3, **0.6**]	2
	sample_r	[1.00, **1.01**]	2
	ens_size	[**50**, 200]	2
	pretr_iter	100	1
	iter	[**300**, 1000]	2
	LR	[1e-2, 1e-3, 1e-4]	3
	wght_dc	0	1
		Total #models =	192

author-recommended values **in bold**.

HP Sensitivity of Deep OD



"I suppose I'll be the one to mention the elephant in the room."

Img src. https://andystrangewayindependent.com/2017/04/06/elephant-in-the-room/

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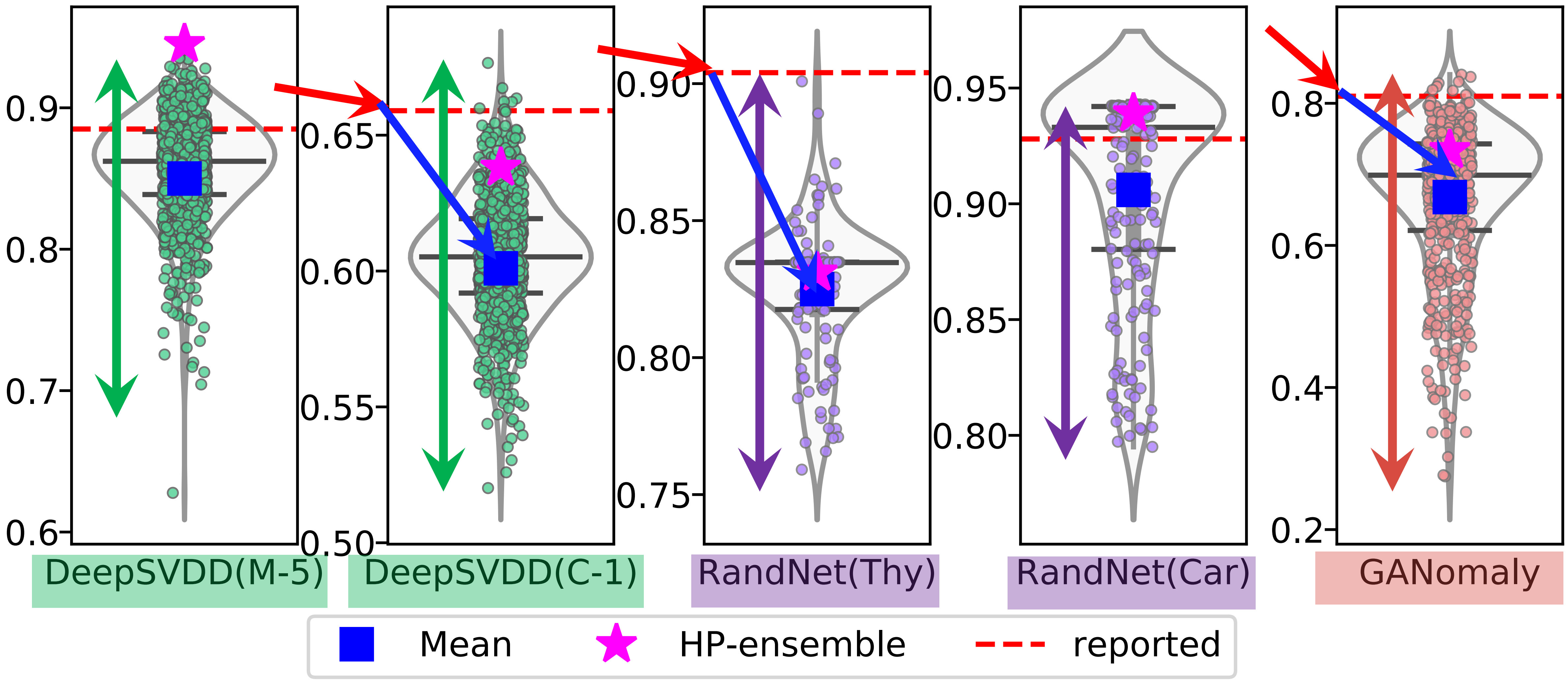

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Our Work

Classical OD

Deep OD

MetaOD

Use HPs that did well on "similar" datasets

+ Zero-shot (no test-time eval)

- Fixed candidate pool

HPOD

+ Continuous space HP search

+ Applies to DL-OD (bigger search space, hard to pre-specify)

ELECT

+ Quantify performance-driven similarity

- Iterative, though not as expensive as ensembles

- Fixed candidate pool

RobOD

Hyper-Ensemble

+ Robust to its own HPs (HP range, #models)

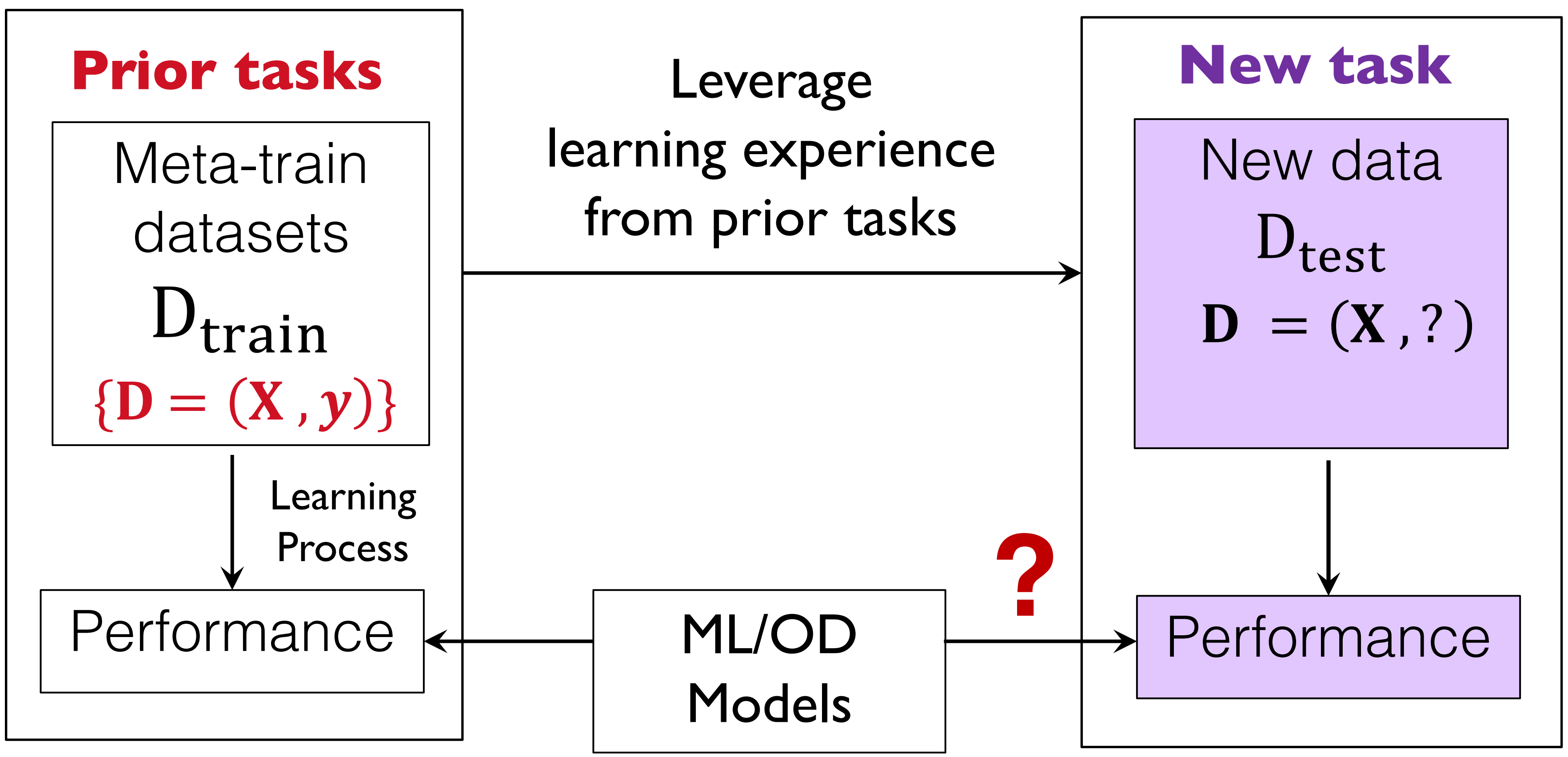
- Relatively slower but new speed-up strategies

HyPer

+ Train one hypernetwork, get many deep OD models

Our Work

Key concept is **meta-learning**: *What can we transfer from historical_OD_tasks_with_labels to a new_task_without_labels?*



Vanschoren, Joaquin. "Meta-learning: A survey." arXiv preprint arXiv:1810.03548 (2018).

Our UOMS Work

MetaOD
https://github.com/yzhao062/MetaOD

ELECT
https://github.com/yzhao062/ELECT

HPOD
https://github.com/yzhao062/HPOD

HyPer
https://github.com/inreview23/HYPER

RobOD
https://anonymous.4open.science/r/9BD1-ROBOD/

Roadmap

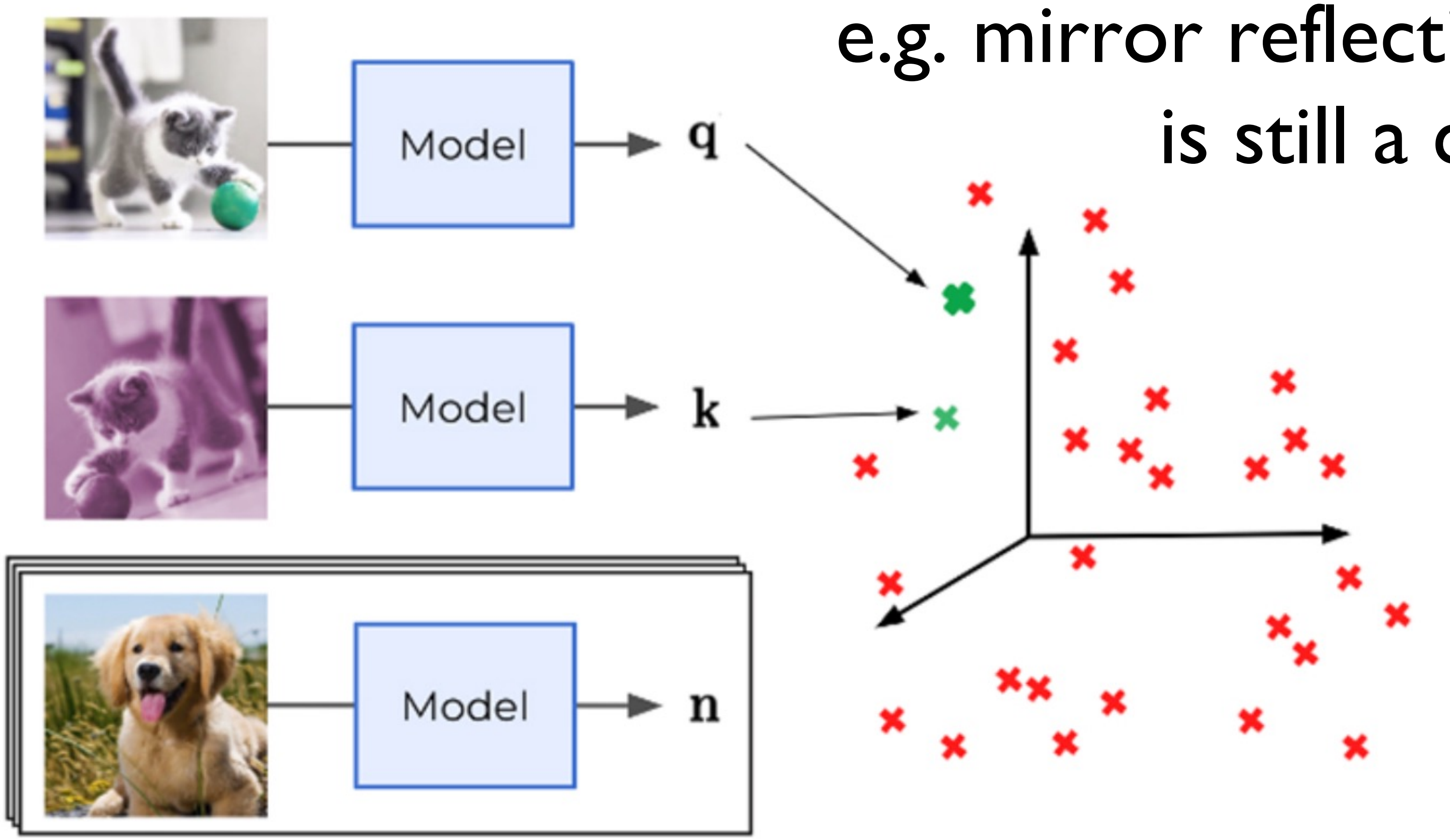
1. Unsupervised Learning
   Hyperparameter sensitivity in OD
2. Self-Supervised (SS) Learning for OD
   toward Self-Tuning SS-OD
3. Zero-shot OD: Model Selection Bygone?

Self-supervised Learning

- Learning without teacher (= self-generated labels)

- **SSL for generalization:**

e.g. mirror reflection of a cat
is still a cat



https://europe.naverlabs.com/blog/improving-self-supervised-representation-learning-by-synthesizing-challenging-negatives/

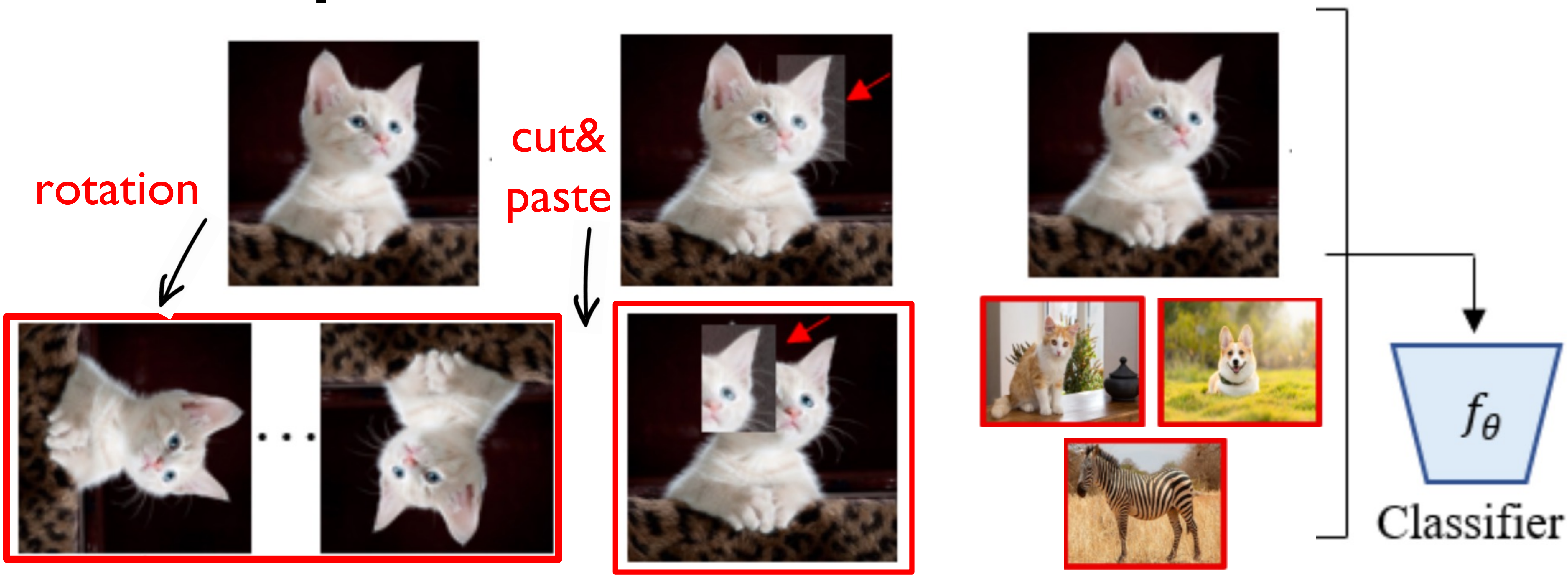
augmentation and contrastive learning

SSL based AD

- SSL for generalization in ML

vs.

- **SSL for pseudo-labels in OD**



auto-generating pseudo-labels (via augmentation) (via outlier exposure)

the Good & the Bad of SSL-based AD

- SSL offers opportunities for many unsupervised AD tasks
- **(Traditional) unsupervised methods:**
  - Assume a **uniform prior** for the anomaly-generating distribution
  - Require a massive number of data to fill in the space
- **SSL for AD:**
  - Create a shift toward various **non-uniform** priors of the anomalies
  - Efficient and effective especially in high-dimensions (e.g., images)

the Good & the Bad of SSL-based AD

- Many different possible augmentation functions:
  - Geometric: Crop (Chen et al., 2020), Rotate, Flip, and GEOM (Golan & El-Yaniv, 2018).
  - Local: CutOut (Devries & Taylor, 2017), CutPaste and CutPaste-scar (Li et al., 2021).
  - Elementwise: Blur (Chen et al., 2020), Noise, and Mask (Vincent et al., 2010).
  - Color-based: Invert and Color (jittering) (Chen et al., 2020).
  - Mixed ("cocktail"): SimCLR (Chen et al., 2020).

- Data augmentation is a hyperparameter (HP):
  - Supervised ML community integrates "data augmentation HPs" into model selection (MacKay et al., 2019; Zoph et al., 2020; Ottoni et al., 2023; Cubuk et al., 2019)…
  - … while the OD community grapples with the choice.

Augmentation Choice Matters

The pseudo abnormality learning uses CutMix [66] to create pseudo anomaly samples on all datasets except the three medical datasets, on which DRA uses external data from another medical dataset LAG [26] as the pseudo anomaly source

Anomaly Category		Augmentation			External	
		CP-Scar	CP-Mix	CutMix	MVTec AD	LAG
Carpet	Color	0.743±0.142	**0.967**±0.048	0.886±0.042	0.615±0.028	0.711±0.041
	Cut	0.853±0.098	0.862±0.072	**0.922**±0.038	0.688±0.019	0.721±0.021
	Hole	0.809±0.033	**0.955**±0.024	0.947±0.016	0.712±0.015	0.823±0.020
	Metal	0.858±0.197	0.840±0.096	**0.933**±0.022	0.764±0.039	0.670±0.037
	Thread	0.987±0.013	0.988±0.011	**0.989**±0.004	0.966±0.003	0.968±0.005
	Mean	0.850±0.070	0.922±0.012	**0.935**±0.013	0.749±0.006	0.779±0.017
AITEX	Broken_end	0.584±0.127	0.750±0.115	0.693±0.099	**0.793**±0.043	0.722±0.072
	Broken_pick	0.616±0.111	0.671±0.082	**0.760**±0.037	0.603±0.017	0.584±0.034
	Cut_selvage	0.676±0.032	0.653±0.091	**0.777**±0.036	0.690±0.013	0.683±0.035
	Fuzzyball	0.639±0.056	0.582±0.067	0.701±0.093	**0.743**±0.053	0.588±0.112
	Nep	0.679±0.060	0.706±0.096	0.750±0.038	**0.774**±0.029	0.739±0.012
	Weft_crack	0.470±0.209	0.507±0.293	**0.717**±0.072	0.671±0.031	0.480±0.140
	Mean	0.611±0.064	0.645±0.070	**0.733**±0.009	0.712±0.010	0.633±0.049
ELPV	Mono	0.665±0.098	0.622±0.067	**0.731**±0.021	0.543±0.064	0.544±0.041
	Poly	0.755±0.006	0.807±0.085	0.800±0.064	0.749±0.052	**0.808**±0.056
	Mean	0.710±0.046	0.715±0.076	**0.766**±0.029	0.646±0.042	0.676±0.031
Hyper-Kvasir	Barretts	0.832±0.016	0.735±0.028	0.761±0.043	**0.834**±0.024	0.824±0.006
	B.-short-seg	0.827±0.054	0.719±0.049	0.695±0.030	**0.839**±0.038	0.835±0.021
	Esophagitis-a	0.832±0.024	0.751±0.023	0.763±0.070	0.811±0.031	**0.881**±0.035
	E.-b-d	0.805±0.035	0.749±0.060	0.782±0.028	**0.847**±0.017	0.837±0.009
	Mean	0.824±0.020	0.739±0.007	0.751±0.021	0.833±0.023	**0.844**±0.009

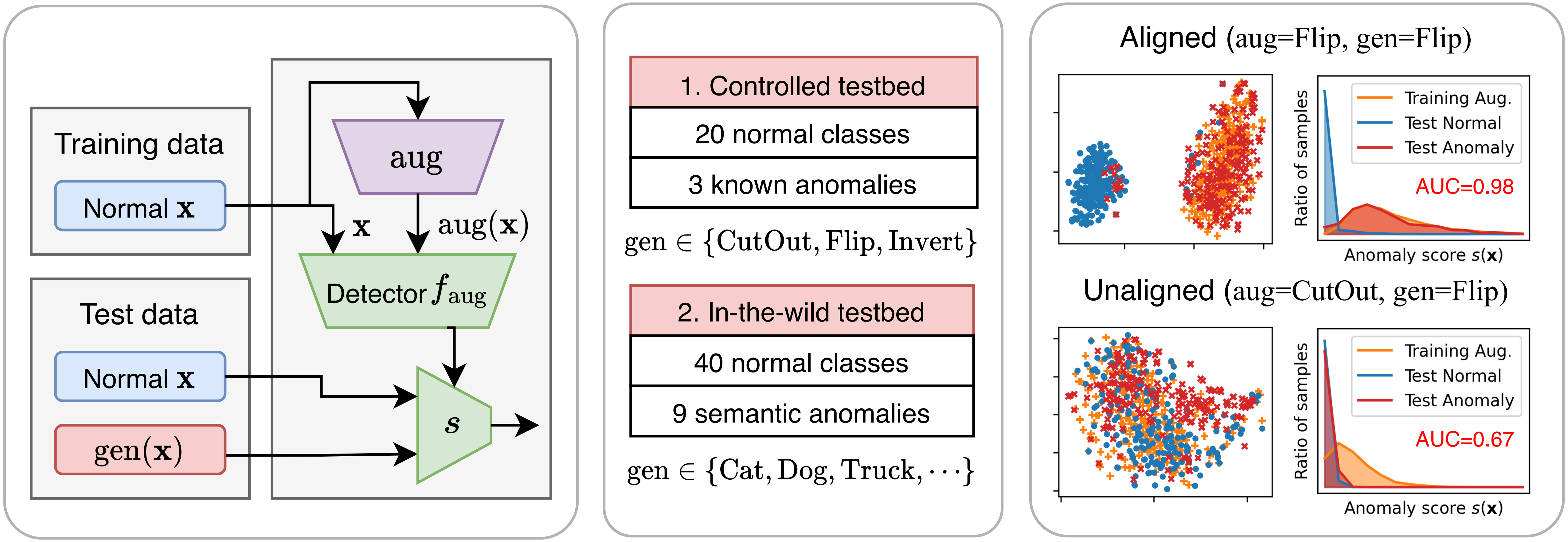
Medical dataset

Best result

Worst result

Ding, Choubo, Guansong Pang, and Chunhua Shen. "Catching Both Gray and Black Swans: Open-set Supervised Anomaly Detection." IEEE/CVF CVPR (2022).

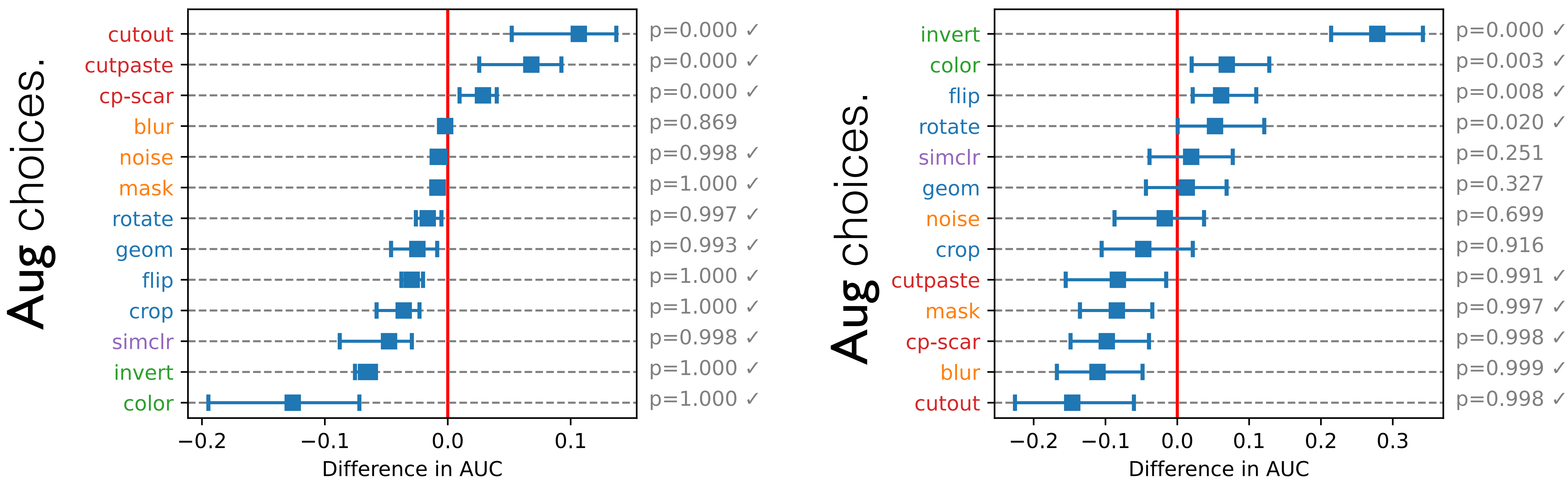
SSL based AD



(a) SSAD Framework (b) Testbeds (c) Main finding: alignment matters

Data Augmentation is a Hyperparameter: Cherry-picked Self-Supervision for Unsupervised Anomaly Detection is Creating the Illusion of Success.
Jaemin Yoo, Tiancheng Zhao, Leman Akoglu.
Transactions on Machine Learning Research (TMLR) (July 2023)

SSL based AD



(a) DAE on **gen**:=CutOut

(b) DeepSAD on **gen**:=Invert

Alignment/agreement (**gen, aug**) matters.

Data Augmentation is a Hyperparameter: Cherry-picked Self-Supervision for Unsupervised Anomaly Detection is Creating the Illusion of Success.

Jaemin Yoo, Tiancheng Zhao, Leman Akoglu.

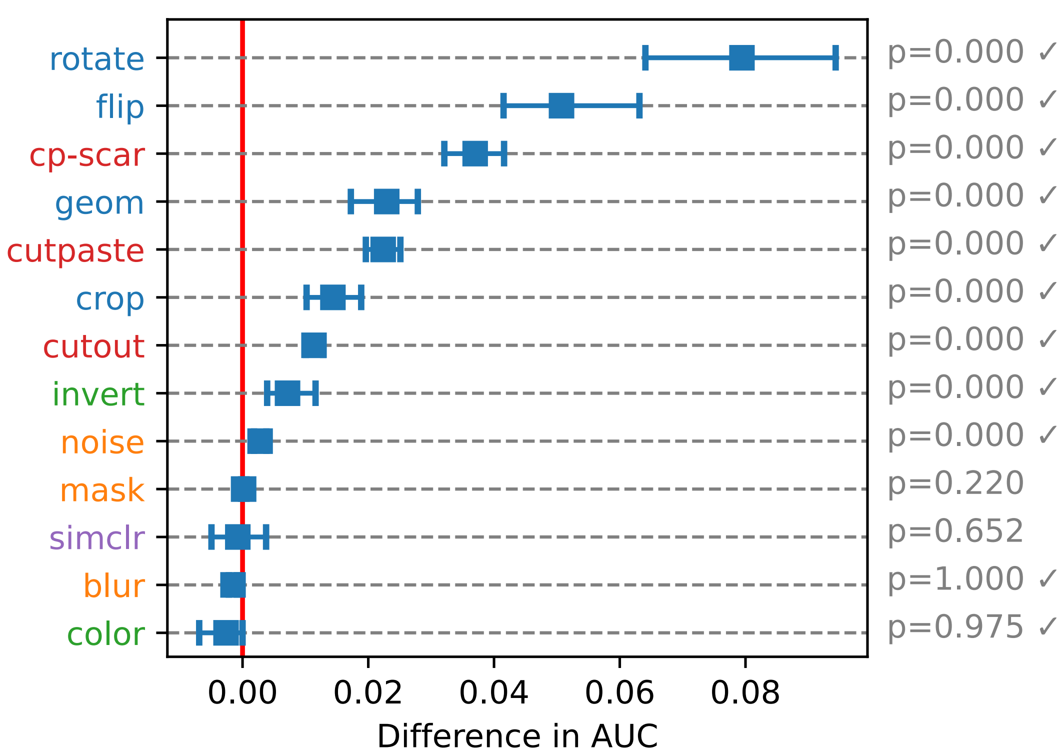
Transactions on Machine Learning Research (TMLR) (July 2023)

SSL based AD

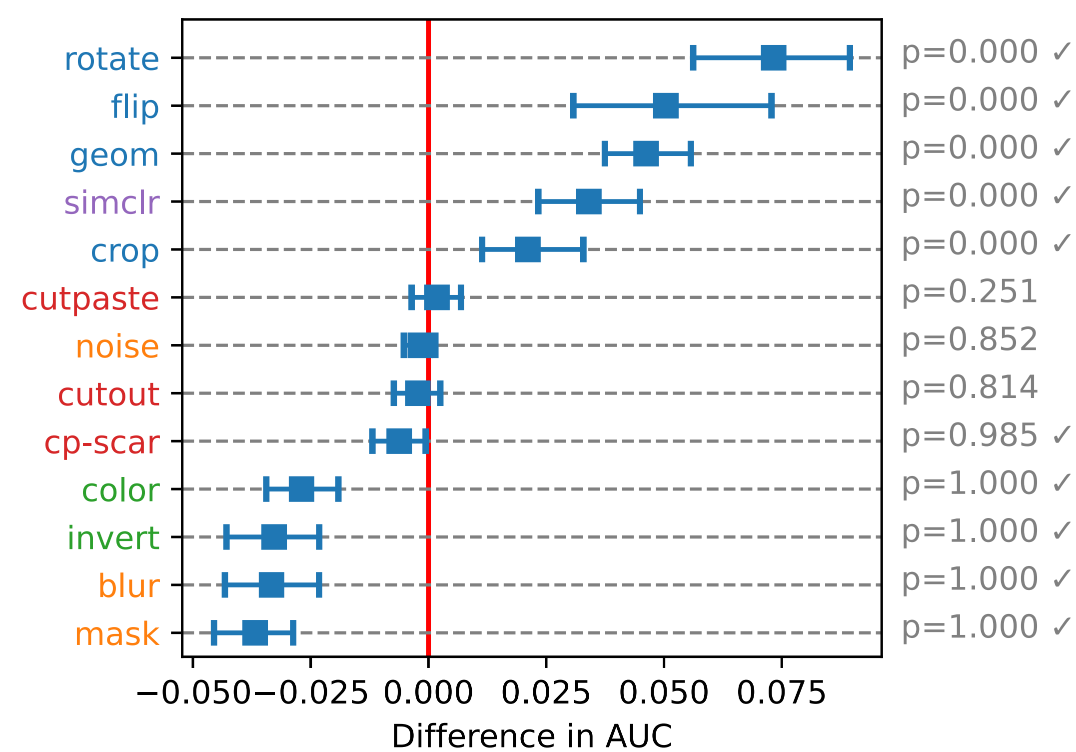
Alignment/agreement (**gen, aug**) matters.

Augment (= aug)	Anomaly-generating function (= gen)											
	DAE				DeepSAD				AP			
	Cut.	Flip	Invert	Sem.	Cut.	Flip	Invert	Sem.	Cut.	Flip	Invert	Sem.
CutOut	**0.974**	0.593	0.654	0.604	**0.999**	0.580	0.556	0.592	**0.797**	0.503	0.508	0.505
Flip	0.771	**0.865**	0.772	**0.691**	0.727	**0.876**	0.731	0.691	0.639	**0.959**	0.836	0.780
Invert	0.746	0.663	**0.970**	0.690	0.674	0.659	**0.973**	**0.695**	0.645	0.717	**0.994**	0.753
GEOM	0.813	0.726	0.724	0.621	0.938	0.758	0.699	0.682	0.760	0.944	0.881	**0.863**

In-the-wild testbed (i.e. gen is unknown):



(a) DAE (vs. AE)



(b) DeepSAD (vs. DeepSVDD)

Roadmap

1. Unsupervised Learning
   Hyperparameter sensitivity in OD

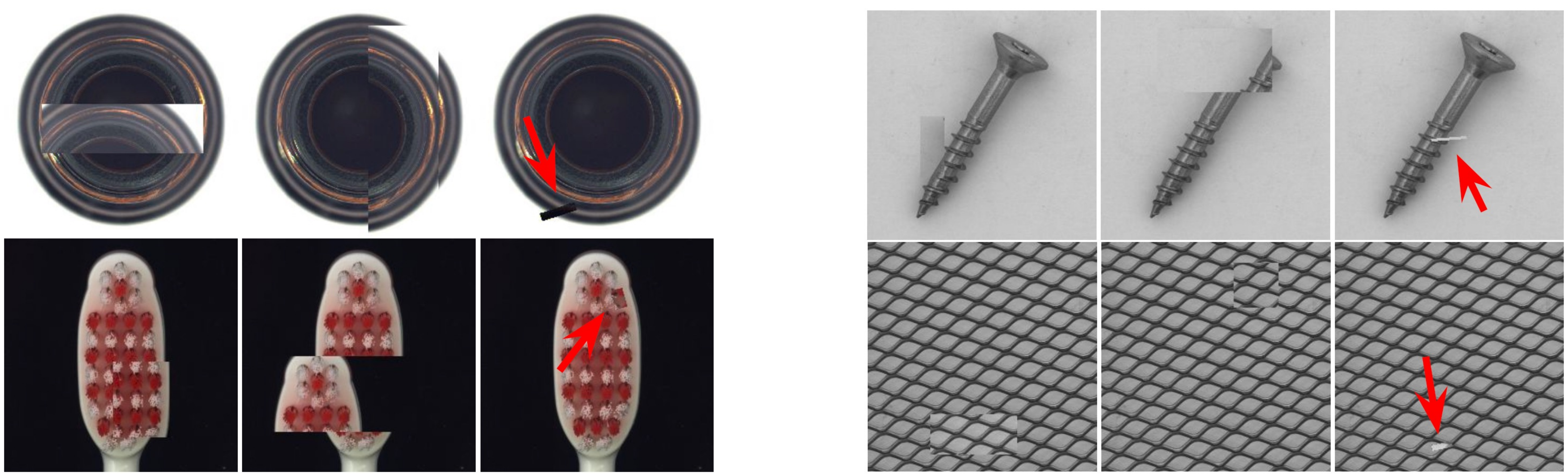
2. Self-Supervised (SS) Learning
   toward Self-Tuning SS-OD

3. Zero-shot OD: Model Selection Bygone?

Self-supervised AD

- **Self-supervised anomaly detection (SSAD)** is a promising direction
  - **Idea:** Generate pseudo anomalies with an augmentation function f_{aug}
- **How SSAD works:**
  - Create $\mathcal{D}_{\text{aug}}$ by applying f_{aug} to normal data $\mathcal{D}_{\text{trn}}$
  - Train a supervised classifier ϕ to classify between $\mathcal{D}_{\text{trn}}$ and $\mathcal{D}_{\text{aug}}$

- **CutPaste** (Li et al., 2021) is an example of f_{aug}

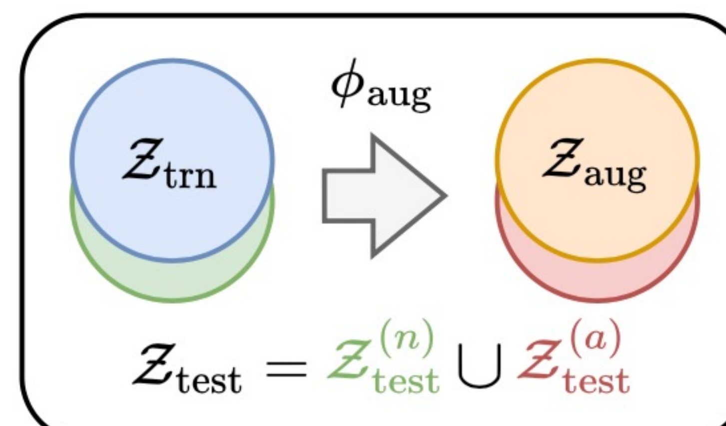


Li et al. "CutPaste: Self-Supervised Learning for Anomaly Detection and Localization." CVPR 2021

Toward Self-Tuning OD

- **Guiding principle**: Transduction
- Vapnik advocated* transductive learning over inductive learning:
  " *One should not solve a more general, harder (intermediate) problem, but rather solve the specific problem at hand directly*. "

- Esp. relevant for operationalizing SSL for AD:
  rather than "imagining" the anomalies, *use test data (w/ unlabeled anomalies) during model training & tuning transductively to measure alignment btwn.* $(\mathcal{Z}_{\text{trn}} \cup \mathcal{Z}_{\text{aug}}, \mathcal{Z}_{\text{test}})$



Unsup. Validation Loss: DSV

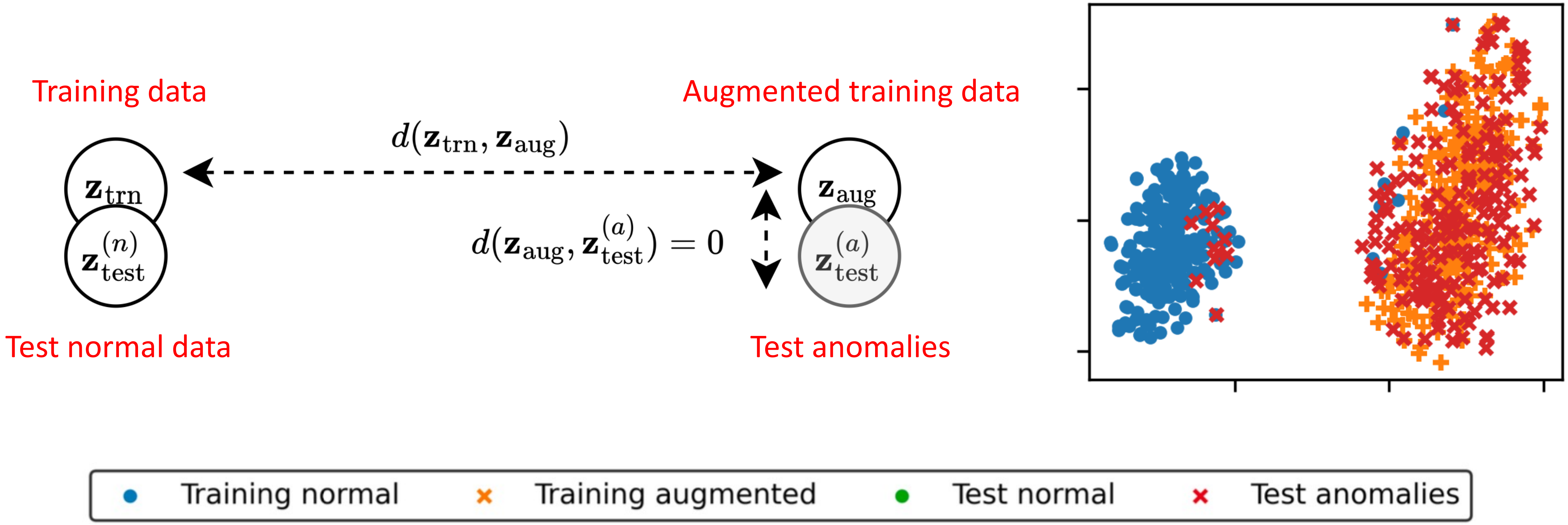
- **Q:** How can we measure the alignment between f_{aug} and f_{gen}?
- **Idea 1:** Measure the distance between embeddings generated by ϕ

$$\boxed{\mathcal{L}_{\text{ali}} = d\left(\mathcal{Z}_{\text{aug}}, \mathcal{Z}_{\text{test}}^{(a)}\right)}$$

  - d: Distance function between sets of vectors
  - $\mathcal{Z}_{\text{aug}}$: Set of embeddings for augmented training data
  - $\mathcal{Z}_{\text{test}}^{(a)}$: Set of embeddings for test anomalies

- **Idea 2:** Decompose it into **discordance** and **separability**
  - They consider two different aspects of the alignment
- **Idea 3:** Design **surrogate losses** to estimate the two terms
  - Design surrogate losses to avoid using any labeled data

Alignment Validation Loss

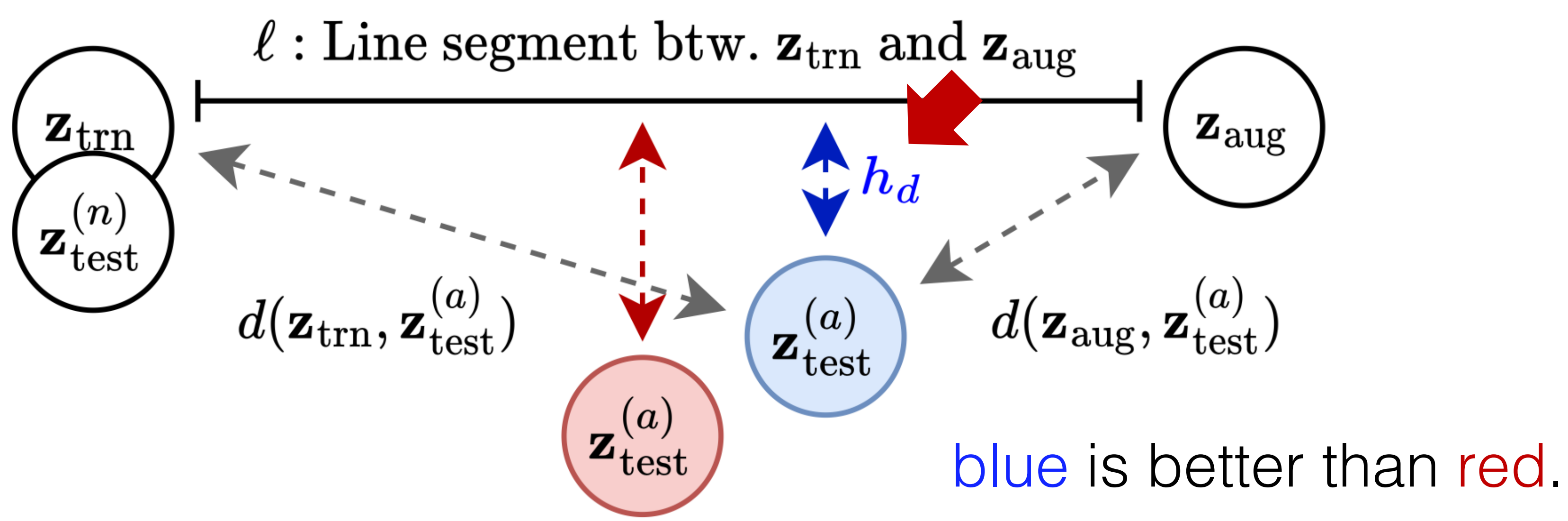


Unsupervised Validation

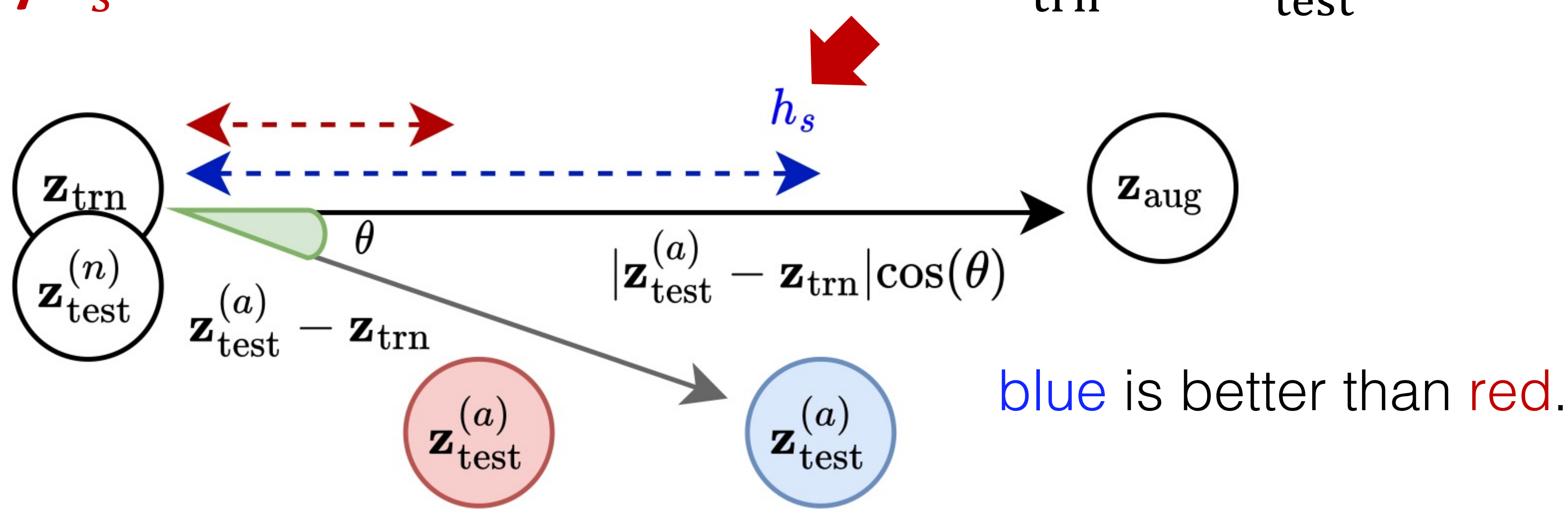
DSV: An Alignment Validation Loss for Self-supervised Outlier Model Selection.
Jaemin Yoo, Yue Zhao, Lingxiao Zhao, Leman Akoglu.
ECML PKDD 2023

Unsup. Validation Loss: DSV

- **Discordance** h_d measures the distance between $\mathcal{Z}_{\text{test}}^{(a)}$ and ℓ

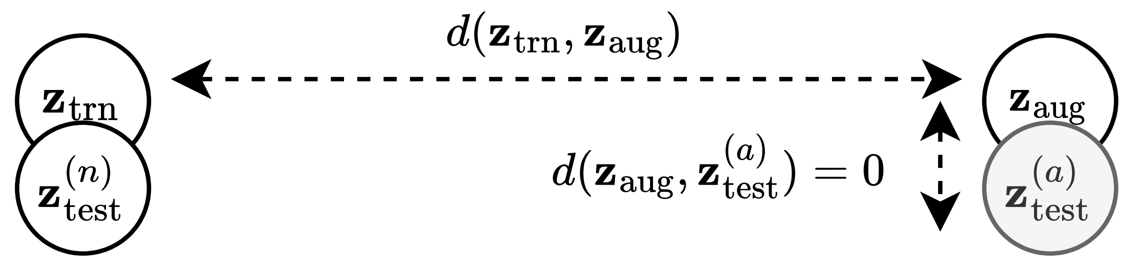


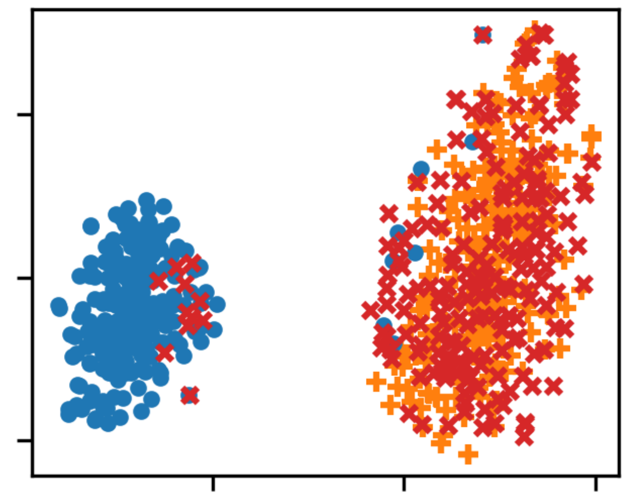
- **Separability** h_s measures the distance between $\mathcal{Z}_{\text{trn}}$ and $\mathcal{Z}_{\text{test}}^{(a)}$ on ℓ



Alignment Validation Loss

(a) Ideal case





WANT:

Low $$\mathcal{L}_{\text{dis}} = \frac{d(\mathcal{Z}_{\text{trn}} \cup \mathcal{Z}_{\text{aug}}, \mathcal{Z}_{\text{test}})}{d(\mathcal{Z}_{\text{trn}}, \mathcal{Z}_{\text{aug}})}$$

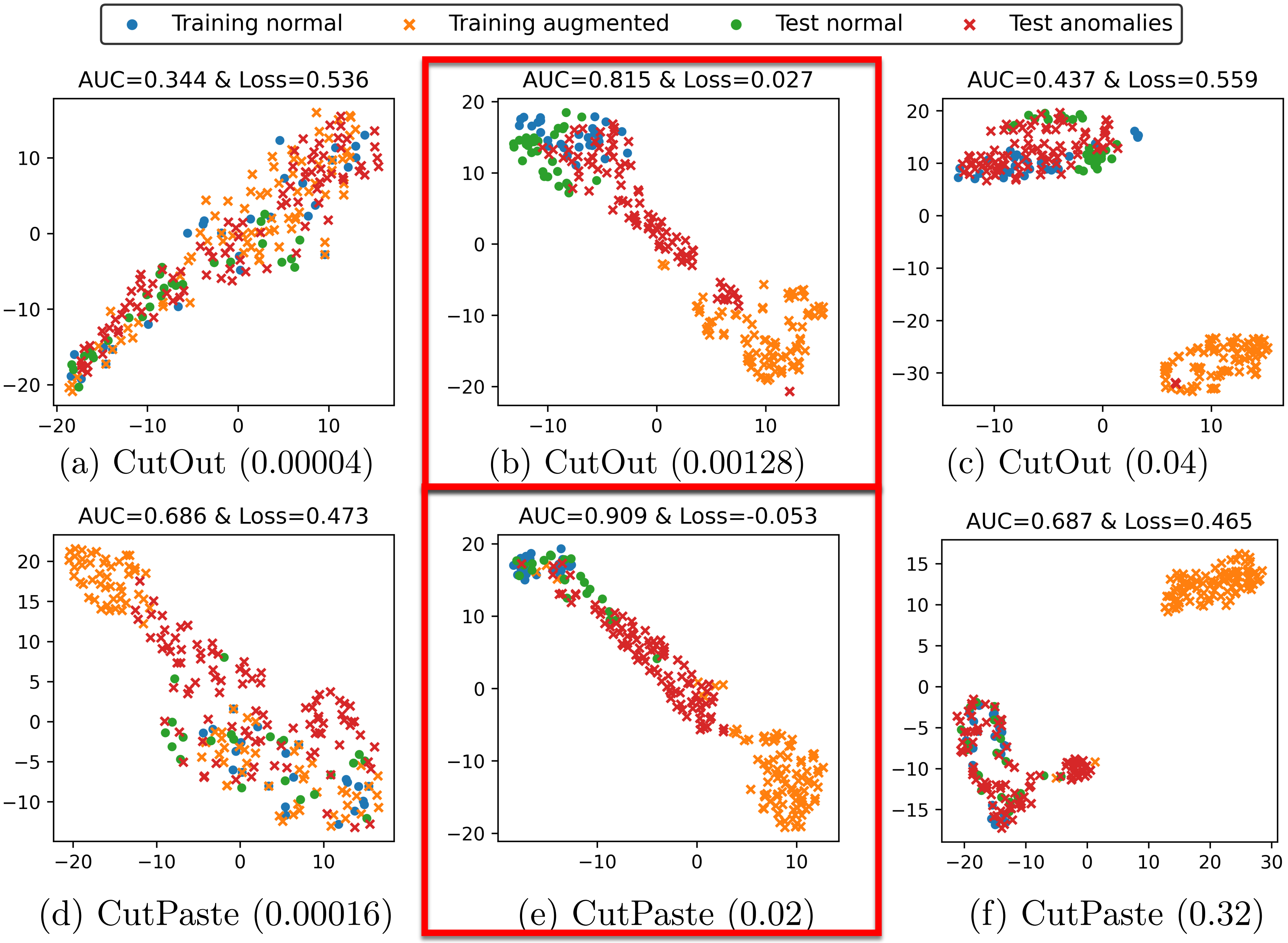
High $$\mathcal{L}_{\text{sep}} = \frac{\text{std}(\{\text{proj}(\mu_{\text{trn}}, \mathbf{z}_{\text{aug}}, \mathbf{z}_{\text{test}})\})}{d(\mathcal{Z}_{\text{trn}}, \mathcal{Z}_{\text{aug}})}$$

Unsupervised Validation

Surrogate losses use the unlabeled test data as a whole.

DSV: An Alignment Validation Loss for Self-supervised Outlier Model Selection.
Jaemin Yoo, Yue Zhao, Lingxiao Zhao, Leman Akoglu.
ECML PKDD 2023

DSV Loss for Model Sel.



DSV: An Alignment Validation Loss for Self-supervised Outlier Model Selection.
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ECML PKDD 2023

Quantitative Evaluation

Table 1: Average AUC (top) and rank (bottom) across 21 different tasks in the two datasets. The best is in bold, and the second best is underlined. Our DSV achieves the best in six, and the second-best in two out of the 8 cases.

f_{aug}	Avg.	Rand.	Base	MMD	STD	MC	SEL	HITS	**DSV**
CutOut	0.739	<u>0.776</u>	0.741	0.735	0.739	0.749	0.727	0.757	**0.813**
CutAvg	0.739	**0.817**	0.721	0.692	0.745	0.751	0.744	0.742	<u>0.806</u>
CutDiff	0.743	0.711	0.739	0.730	0.744	0.747	0.741	<u>0.777</u>	**0.811**
CutPaste	0.788	0.841	0.694	0.756	0.818	<u>0.862</u>	0.830	0.850	**0.884**

f_{aug}	Avg.	Rand.	Base	MMD	STD	MC	SEL	HITS	**DSV**
CutOut	7.33	6.10	6.62	6.93	6.29	6.50	7.10	<u>5.43</u>	**3.79**
CutAvg	7.00	<u>5.02</u>	7.64	8.36	5.52	5.48	5.98	5.60	**4.19**
CutDiff	6.43	7.24	6.45	7.38	6.00	<u>5.64</u>	6.24	6.21	**3.60**
CutPaste	7.67	6.29	8.67	7.21	5.60	**4.33**	5.17	4.64	<u>4.57</u>

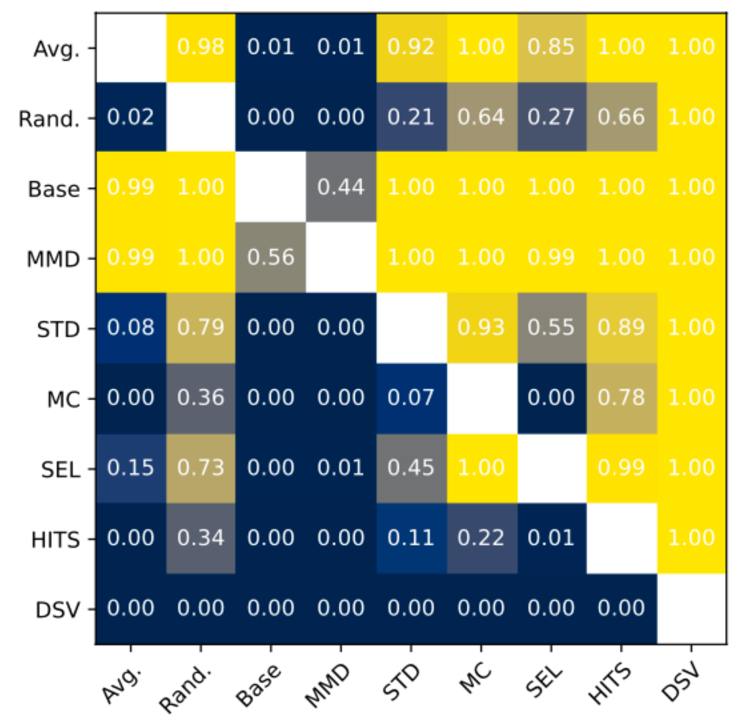


Fig. 5: Wilcoxon signed rank test for all pairs of approaches. DSV is superior to all other approaches with p-values smaller than 0.001.

DSV: An Alignment Validation Loss for Self-supervised Outlier Model Selection.
Jaemin Yoo, Yue Zhao, Lingxiao Zhao, Leman Akoglu.
ECML PKDD 2023

Roadmap

1. Unsupervised Learning
   Hyperparameter sensitivity in OD

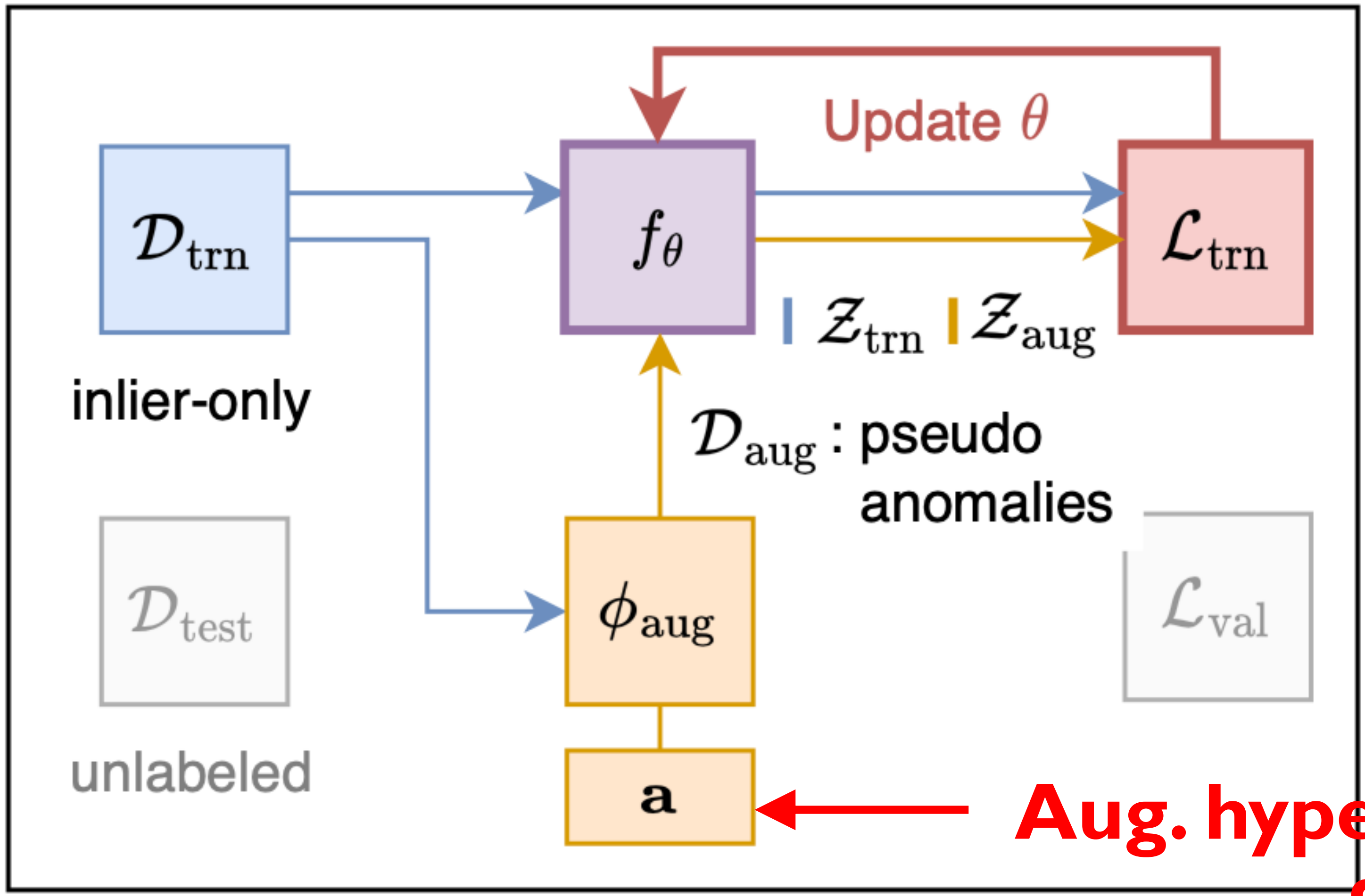
2. Self-Supervised (SS) Learning
   toward Self-Tuning SS-OD
   - DSV **Paper and code:** https://github.com/jaeminyoo/DSV
   - <u>End-to-end Self-Tuning</u>
     vs. DSV: <u>not</u> differentiable

3. Zero-shot OD: Model Selection Bygone?

End-to-end Self-tuning SSAD

Train **detector** f_θ given aug-HPs **a (fixed)**

based on training loss

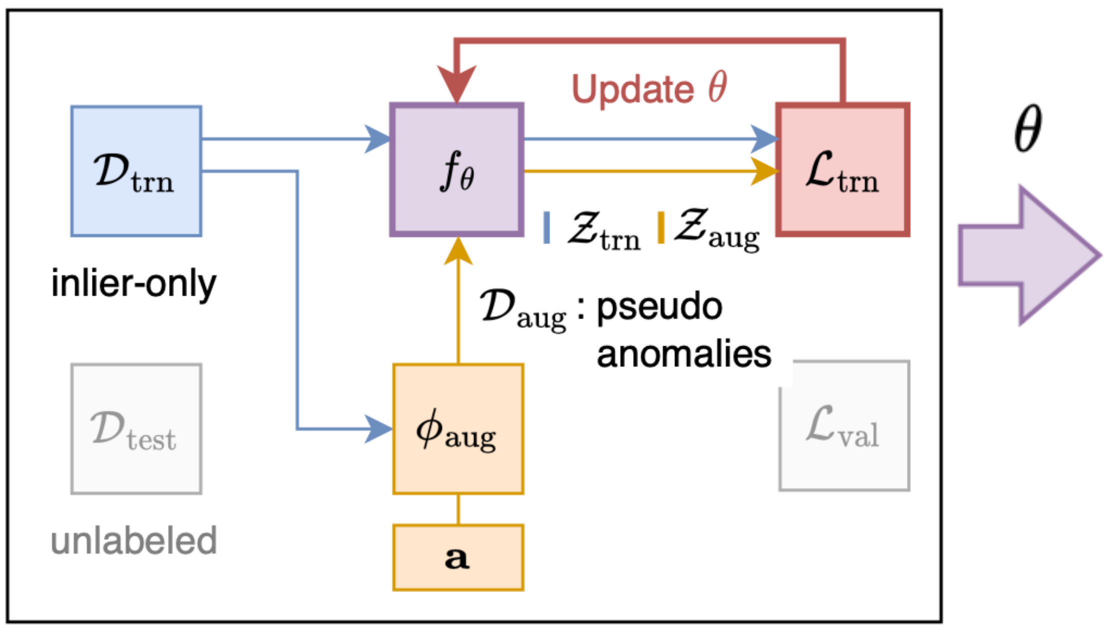


Aug. hyperparameters fixed

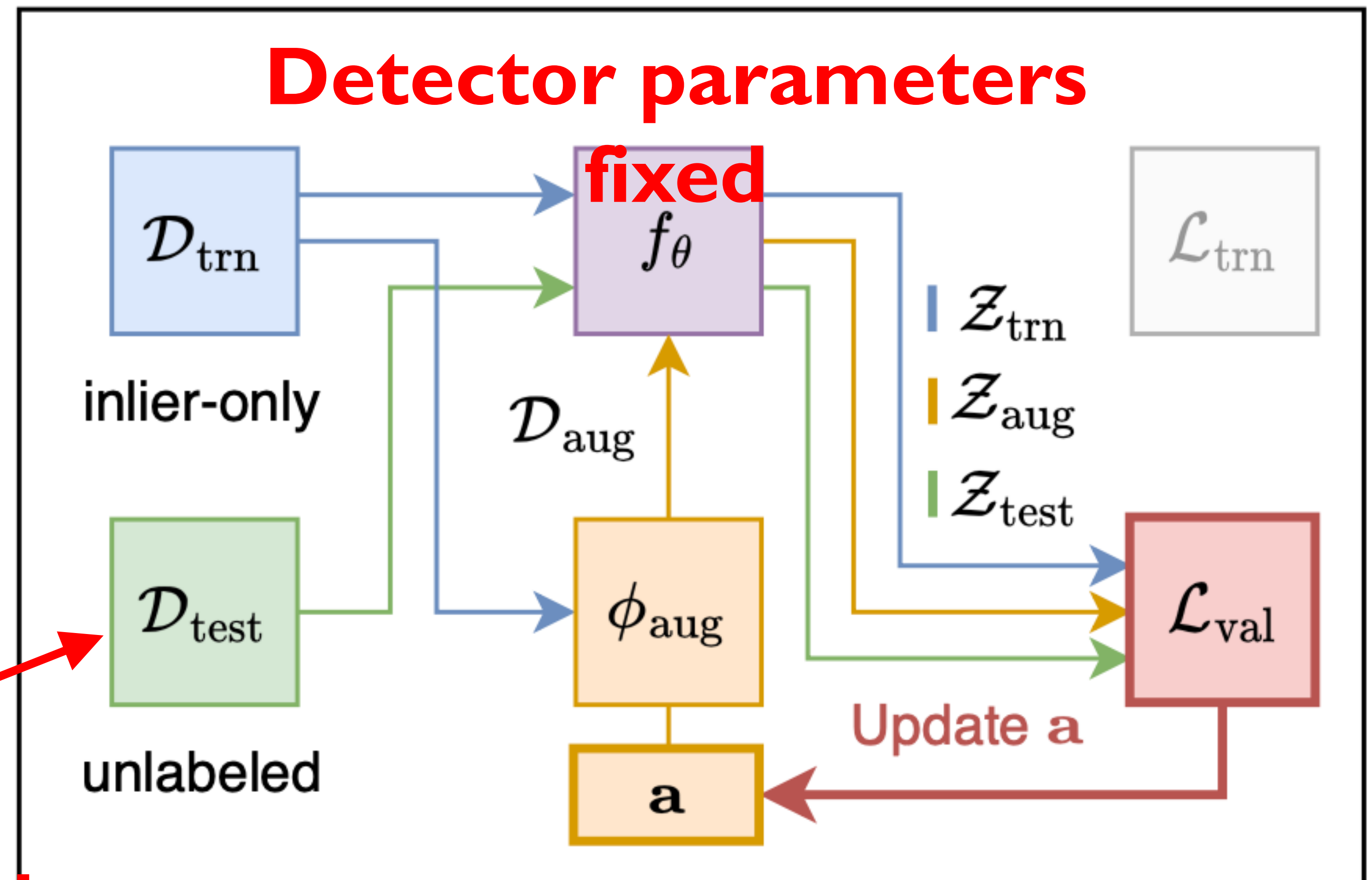
Self-Tuning Self-Supervised Anomaly Detection.
Jaemin Yoo, Lingxiao Zhao, Leman Akoglu.
arXiv:2306.12033 (2025)

End-to-end Self-tuning SSAD

Train detector f_θ given $\mathbf{a}$
based on training loss



Update $\mathbf{a}$ given fixed f_θ
based on validation loss



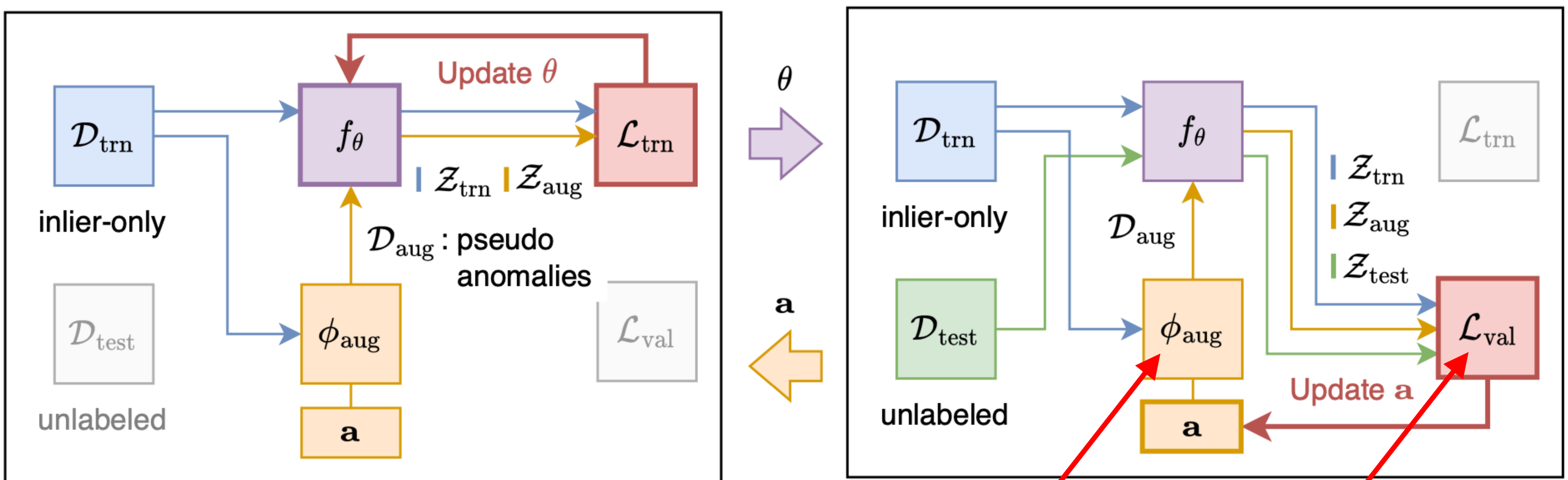
Transductive
w/ unlabeled test data

Self-Tuning Self-Supervised Anomaly Detection.
Jaemin Yoo, Lingxiao Zhao, Leman Akoglu.
arXiv:2306.12033 (2025)

End-to-end Self-tuning SSAD

Training detector f_θ given $\mathbf{a}$
based on training loss

Updating $\mathbf{a}$ given fixed f_θ
based on validation loss



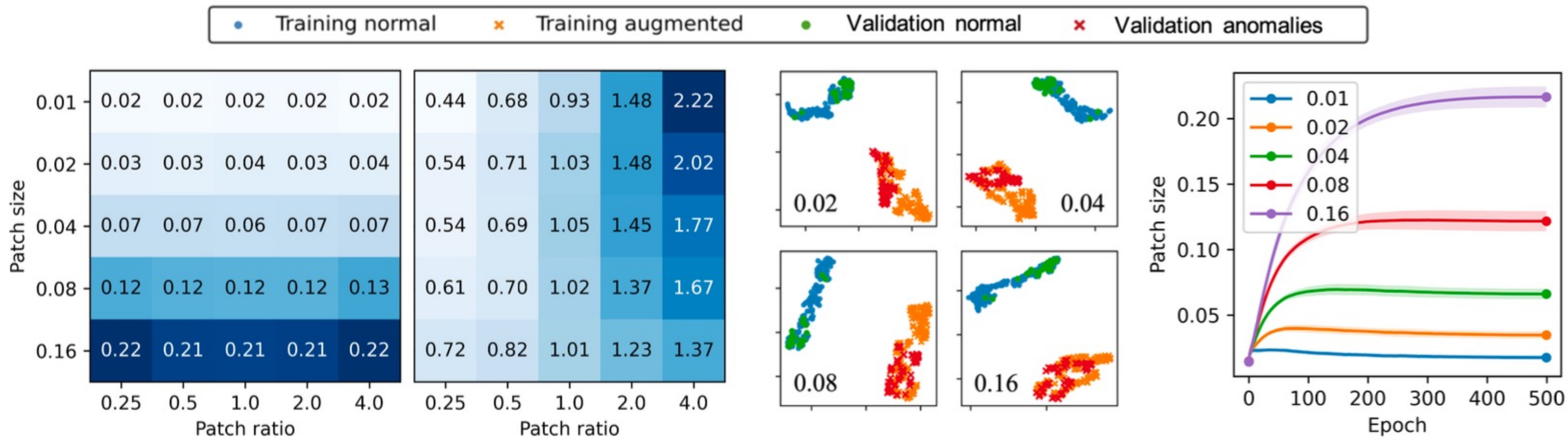
Two key components:

1) Differentiable augmentation

2) Differentiable unsup. val. loss

Self-Tuning Self-Supervised Anomaly Detection.
Jaemin Yoo, Lingxiao Zhao, Leman Akoglu.
arXiv:2306.12033 (2025)

Self-tuning HPs – Eval.



(a) Learned values of patch size (left) & ratio (right) **(b) *t*-SNE embeddings** **(c) Patch size during training**

Figure 4: Experimental results on demonstrative examples, where we create 25 types of anomalies using CutDiff with five different patch size and patch ratio values, respectively. (a) Each cell's value represents what ST-SSAD has learned through end-to-end optimization for the (left) patch size and (right) ratio, respectively. ST-SSAD successfully mimics the true values of the patch ratio and size shown in the x- and y-axes, respectively, until (b) the embedding distributions are matched between $\mathcal{Z}_{\text{trn}} \cup \mathcal{Z}_{\text{aug}}$ and $\mathcal{Z}_{\text{val}}$, showing a learning trajectory like (c). Note that AUC is 1.00 in all 25 tasks.

Self-tuning SSAD – Eval.

Gross/global anomalies: Semantic class as anomaly

Object	Anomaly	AE	D-SVDD	RS-RO	RD-RO	ST-SSAD
Digit 2	Digit 0	0.602	0.472	0.672	0.734	**0.816**
Digit 2	Digit 1	0.544	0.499	0.601	0.609	**0.743**
Digit 2	Digit 3	0.604	0.503	0.664	0.730	**0.832**
Digit 2	Digit 4	0.561	0.511	0.679	0.746	**0.790**
Digit 2	Digit 5	0.625	0.502	0.709	0.824	**0.877**
Digit 2	Digit 6	0.616	0.496	0.726	0.826	**0.887**
Digit 2	Digit 7	0.541	0.496	0.584	0.639	**0.823**
Digit 2	Digit 8	0.616	0.498	0.673	0.738	**0.805**
Digit 2	Digit 9	0.588	0.485	0.625	**0.687**	0.659
Digit 6	Digit 0	0.531	0.480	0.586	0.606	**0.777**
Digit 6	Digit 1	0.517	0.498	0.621	0.610	**0.854**
Digit 6	Digit 2	0.594	0.503	0.735	0.807	**0.916**
Digit 6	Digit 3	0.570	0.507	0.686	0.750	**0.823**
Digit 6	Digit 4	0.525	0.508	0.659	0.703	**0.709**
Digit 6	Digit 5	0.544	0.502	0.615	**0.661**	0.658
Digit 6	Digit 7	0.567	0.505	0.699	0.729	**0.861**
Digit 6	Digit 8	0.546	0.500	0.575	0.641	**0.732**
Digit 6	Digit 9	0.579	0.495	0.708	0.817	**0.944**
p-value		.0000	.0000	.0000	.0000	**Ours**

Self-tuning SSAD – Eval.

Subtle/local anomalies: Industrial defects

Object	Anomaly Type	AE	D-SVDD	RS-CO	RD-CO	RS-CP	RD-CP	RS-CD	RD-CD	**ST-SSAD**
Cable	Bent wire	0.515	0.432	0.556	0.560	0.703	**0.756**	0.527	0.580	0.490
Cable	Cable swap	0.639	0.295	0.483	0.625	0.618	0.683	0.574	**0.696**	0.532
Cable	Combined	0.584	0.587	0.879	0.857	0.880	**0.949**	0.901	0.879	0.925
Cable	Cut inner insulation	0.758	0.591	0.630	0.737	0.766	**0.833**	0.623	0.732	0.667
Cable	Cut outer insulation	**0.989**	0.343	0.695	0.815	0.787	0.871	0.703	0.790	0.516
Cable	Missing cable	0.920	0.466	0.953	0.961	0.755	0.801	0.935	0.945	**0.998**
Cable	Missing wire	0.433	0.494	0.781	0.655	0.501	0.546	0.708	0.620	**0.863**
Cable	Poke insulation	0.287	0.471	0.469	0.527	0.645	**0.672**	0.489	0.503	0.630
Carpet	Color	0.578	0.716	0.669	0.508	0.412	0.287	0.643	0.639	**0.938**
Carpet	Cut	0.198	0.758	0.439	0.608	0.403	0.411	0.490	0.767	**0.790**
Carpet	Hole	0.626	0.676	0.379	0.613	0.404	0.389	0.470	**0.765**	0.590
Carpet	Metal contamination	0.056	**0.739**	0.198	0.304	0.240	0.167	0.255	0.474	0.076
Carpet	Thread	0.394	**0.742**	0.494	0.585	0.469	0.517	0.508	0.679	0.483
Grid	Bent	**0.849**	0.168	0.456	0.322	0.421	0.433	0.337	0.354	0.771
Grid	Broken	0.806	0.183	0.397	0.312	0.487	0.502	0.340	0.392	**0.869**
Grid	Glue	0.704	0.143	0.634	0.568	0.674	0.732	0.681	0.578	**0.906**
Grid	Metal contamination	0.851	0.229	0.421	0.380	0.499	0.514	0.425	0.613	**0.858**
Grid	Thread	0.583	0.209	0.612	0.494	0.500	0.549	0.654	0.611	**0.973**
Tile	Crack	0.770	0.728	0.872	0.993	0.743	0.636	0.837	**0.999**	0.749
Tile	Glue strip	0.697	0.509	0.693	**0.836**	0.665	0.700	0.675	0.831	0.767
Tile	Gray stroke	0.637	0.785	0.845	0.642	0.583	0.657	0.856	0.802	**0.974**
Tile	Oil	0.414	0.690	0.708	0.745	0.464	0.576	0.683	**0.837**	0.554
Tile	Rough	0.724	0.387	0.606	**0.725**	0.631	0.661	0.568	0.657	0.690
	p-value	.0000	.0000	.0000	.0012	.0000	.0000	.0000	.0728	**Ours**

Self-tuning SSAD – Eval.

Case Studies

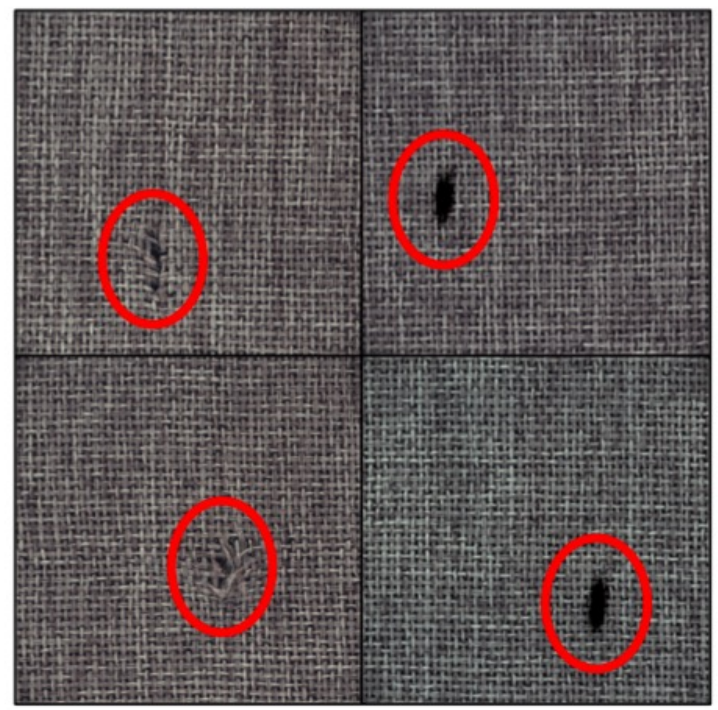
(a) Anomaly (poke_insulation) | Aug. (scale=0.001) | (b) Anomaly (missing_wire) | Aug. (scale=0.081) | (c) Anomaly (missing_cable) | Aug. (scale=0.177)

Effectively learns **cut-diff** size

Self-tuning SSAD – Eval.

Case Studies

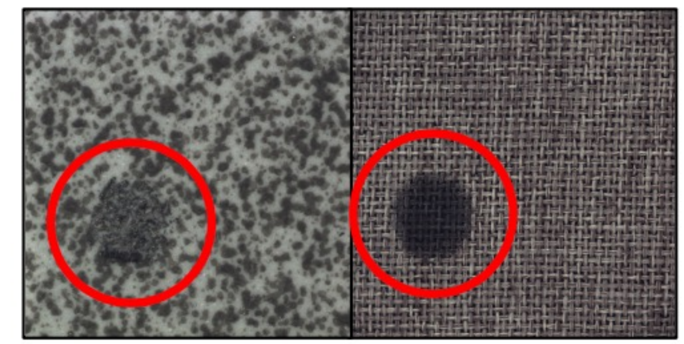
Effectively learns **rotation** angle



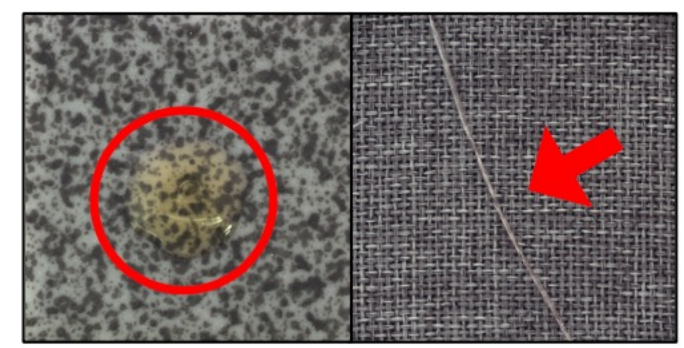
(d) Anomaly (cut) Aug. (scale=0.047)

Effectively learns **cut-diff** size & aspect ratio

(a) Can be found by CutDiff



(b) Difficult with CutDiff



Limited to a few differentiable augmentations

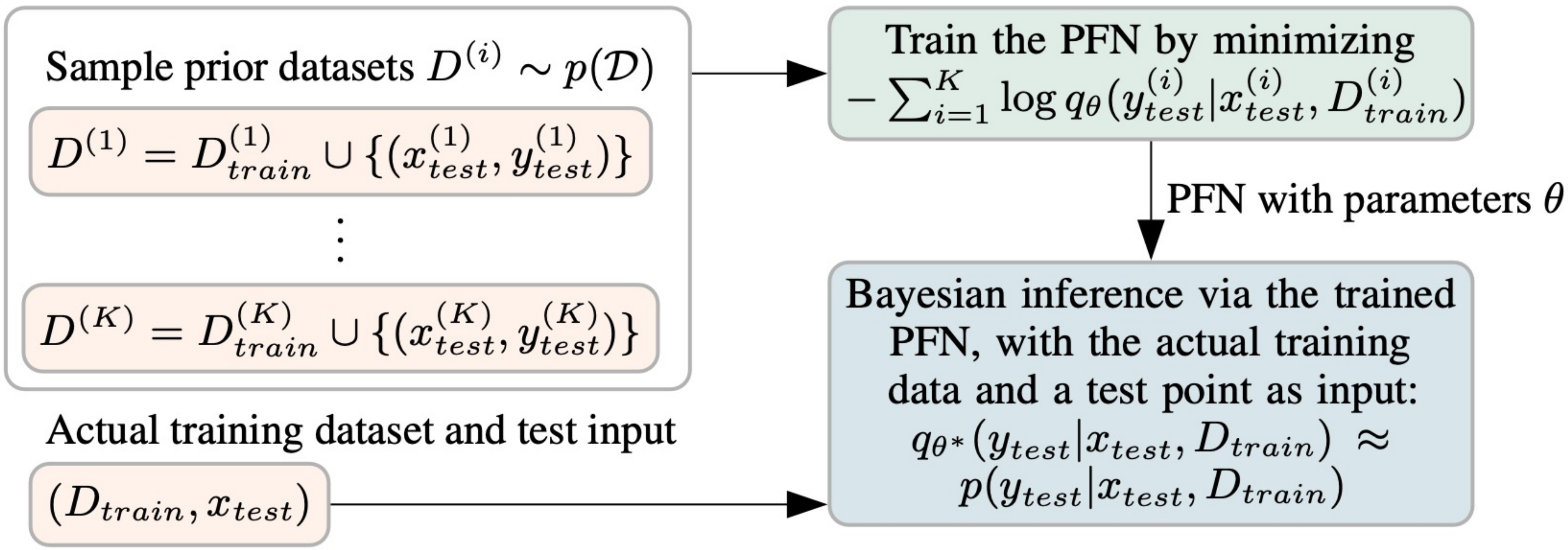
Roadmap

1. Unsupervised Learning
   Hyperparameter sensitivity in OD

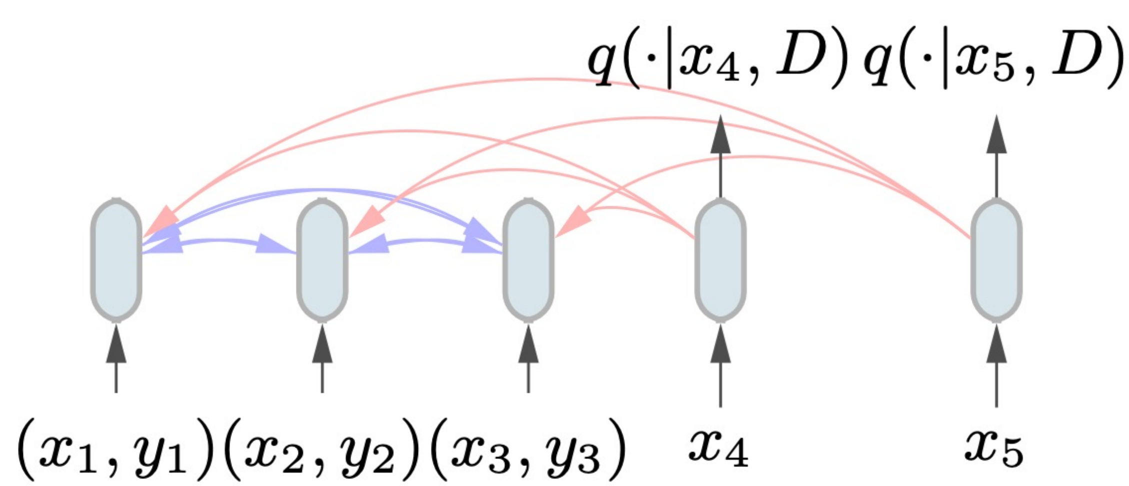
2. Self-Supervised (SS) Learning
   toward Self-Tuning SS-OD

Zero-shot OD: Model Selection Bygone?

PFNs: Prior-data Fitted Networks



Samuel Müller, Noah Hollmann, Sebastian Pineda-Arango, Josif Grabocka, and Frank Hutter. Transformers Can Do Bayesian Inference. ICLR, 2022.

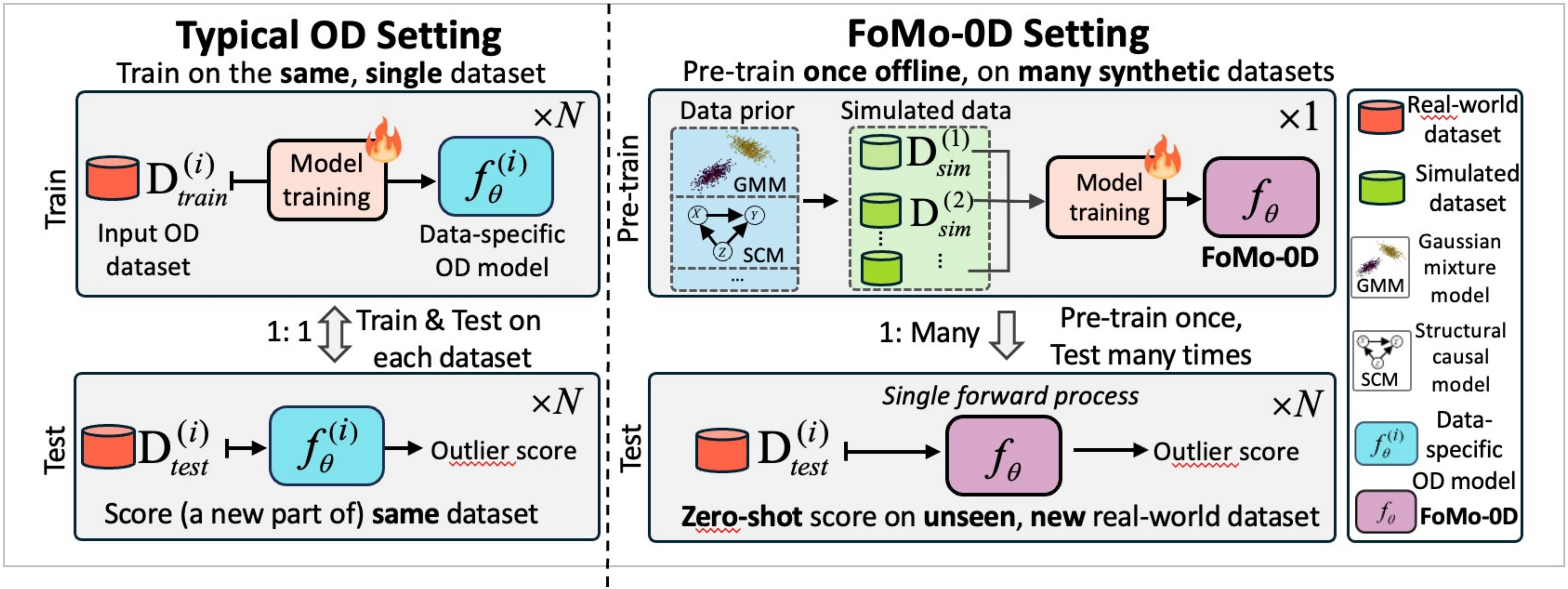


Note:
sample-to-sample
attention

Noah Hollmann, Samuel Müller, Katharina Eggensperger, and Frank Hutter. TabPFN: A Transformer That Solves Small Tabular Classification Problems in a Second. ICLR, 2023.

PFN for OD

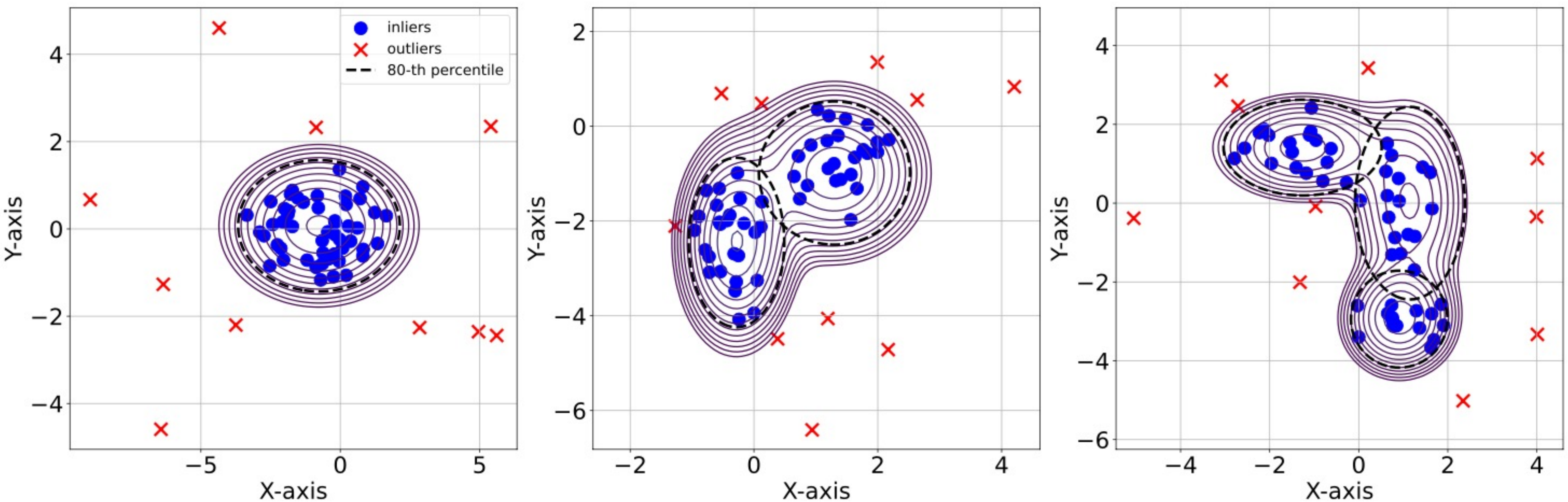
- A **Fo**undation **Mo**del for **0**-shot O**D**



Zero-shot Outlier Detection via Prior-data Fitted Networks: Model Selection Bygone!
Yuchen Shen, Haomin Wen, Leman Akoglu.
Preprint, https://tinyurl.com/4nztsh5t (Sep. 2024)

A very basic data prior: GMMs

- Inliers: GMMs w/ $\leq$5 components w/ $\leq$100 dim.s
  Outliers: Subspace outliers with "inflated" covariance (x5)



30 baselines on 57 real-world datasets:

Table 1: Comparison of methods across datasets. (top row) Rank w.r.t. AUROC performance avg.'ed over 57 datasets is presented for FoMo-0D (with $D = 100$), **top-10 baselines** with default HPs, and **top-4**[7] baselines with performance **avg.**'ed over varying HPs (denoted w/ $^{\text{avg}}$); followed by p-values of the pairwise Wilcoxon signed rank test, comparing FoMo-0D to each baseline (from top to bottom) over All (57) datasets, those (42) w/ $d \leq 100$ and (46) w/ $d \leq 500$ dimensions. FoMo-0D performs as well as **(i.e., statistically no different from) the 2nd best model** (kNN, w/ $p = 0.106$) across All datasets, while it is **comparable to** ($p > 0.05$) **or better than** ($p > 0.95$) **all baselines** over datasets w/ $d \leq 100$ (aligned w/ pretraining where $D = 100$) *and* $d \leq 500$ (generalizing beyond pretraining).

	FoMo-0D	DTE-NP	kNN	ICL	DTE-C	LOF	CBLOF	Feat.Bag.	SLAD	DDPM	OCSVM	DTE-NP$^{\text{avg}}$	kNN$^{\text{avg}}$	ICL$^{\text{avg}}$	DTE-C$^{\text{avg}}$
Rank(avg)	11.886	7.553	9.018	10.851	11.36	12.316	13.342	13.386	12.982	14.061	13.851	9.079	11.105	12.991	22.263
All	-	0.016	0.106	0.462	0.454	0.585	0.750	0.823	0.759	0.901	0.895	0.112	0.315	0.670	1.000
$d \leq 100$	-	0.415	0.700	0.949	**0.953**	**0.970**	**0.971**	**0.996**	0.876	**0.980**	**0.978**	0.752	0.860	**0.958**	**1.000**
$d \leq 500$	-	0.220	0.569	0.827	0.894	**0.960**	**0.968**	**0.994**	0.910	**0.960**	**0.979**	0.607	0.756	0.846	**1.000**

- Statistically no different from 2$^{\text{nd}}$ best baseline (among 30),
- Significantly outperform majority of baselines (p > 0.95)

30 baselines on 57 real-world datasets:

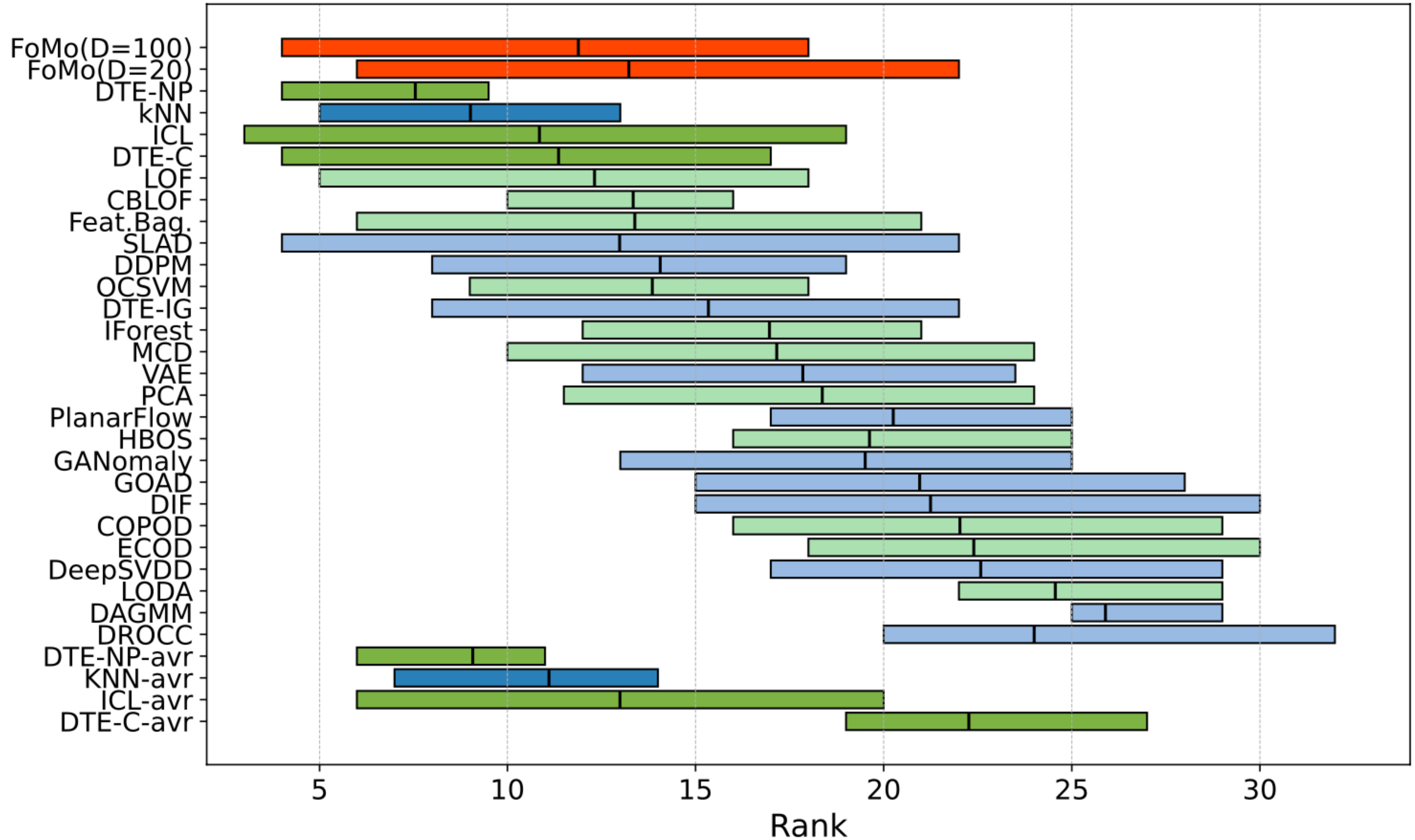
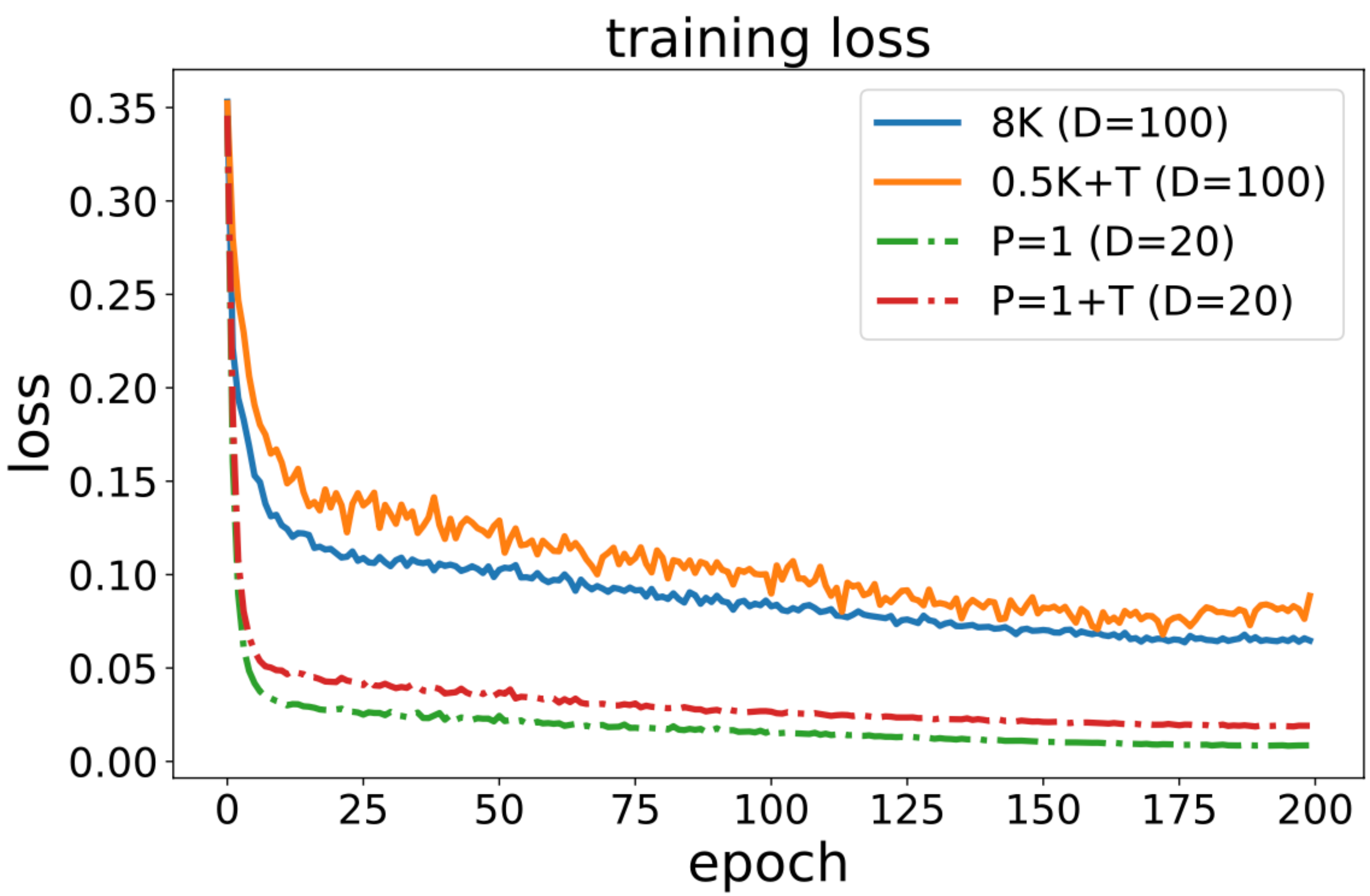


Figure 3: (best in color) Rank (w.r.t. AUROC performance, lower is better) distribution across datasets shown via boxplots for (from top to bottom) FoMo-0D in red, all 26 baselines as ordered by mean p-value[7] (shallow and deep baselines resp. in green and blue), and top-4 baselines' ^{avg} variants.

How far can we go?

- With an **even simpler** data prior; 2 simplifications:
  1) GMMs w/ ≤ 20 dim.s, 2) full-space outliers



200 epochs, 1000 steps each with a batch size of 8 = 1,600,000 pretraining datasets

30 baselines on 57 real-world datasets:

	FoMo-0D	DTE-NP	kNN	ICL	DTE-C	LOF	CBLOF	Feat.Bag.	SLAD	DDPM	OCSVM	DTE-NP^{avg}	kNN^{avg}	ICL^{avg}	DTE-C^{avg}
Rank(avg)	12.59	7.19	8.57	10.34	10.79	11.82	12.81	12.8	12.52	13.50	13.34	8.60	10.63	12.44	21.43
All	-	<u>0.001</u>	<u>0.019</u>	0.089	0.159	0.394	0.434	0.703	0.516	0.752	0.679	<u>0.007</u>	0.062	0.437	1.0
$d \leq 20$	-	0.572	0.789	0.968	0.616	**0.993**	**0.989**	**1.0**	**0.978**	0.906	**0.992**	0.813	0.924	**0.999**	**1.0**
$d \leq 50$	-	0.347	0.794	0.893	0.946	**0.997**	**0.988**	**1.0**	**0.963**	**0.994**	**0.986**	0.574	0.847	**0.995**	**1.0**

Table 2: Comparison of methods across datasets. (top row) Rank w.r.t. AUROC performance avg.'ed over 57 datasets is presented for FoMo-0D (with $D = 20$), **top-10** baselines with default HPs, and **top-4**[7] baselines with performance **avg.**'ed over varying HPs (denoted w/ ^{avg}); followed by p-values of the pairwise Wilcoxon signed rank test, comparing FoMo-0D to each baseline (from top to bottom) over All (57) datasets, those (24) w/ $d \leq 20$ and (38) datasets w/ $d \leq 50$ dimensions, respectively. Even with small $D = 20$, FoMo-0D performs as well as (i.e., statistically no different at 0.05 from) *the top 3rd baseline* (ICL, w/ $p = 0.089$) across All datasets, while it *outperforms the top 5th (LOF) and onward baselines* over datasets w/ $d \leq 20$ (aligned w/ pretraining where $D = 20$) *and* $d \leq 50$

- Statistically no different from 3rd best baseline (among 30),
- Significantly outperform majority of baselines (p > 0.95)

More in the paper…

- Scalable sample-to-sample attention via “routers”
- Scalable data synthesis via linear transformations
- Many ablations
  - Is a more complex data prior preferable?
  - Impact of context size
  - Pretraining data size and diversity
  - Inference time and speed up
  - …

Zero-shot Outlier Detection via Prior-data Fitted Networks: Model Selection Bygone!

Yuchen Shen, Haomin Wen, Leman Akoglu.

Preprint, https://tinyurl.com/4nztsh5t (Sep. 2024)

Take-aways:

A key obstacle for OD in the real world,
especially with Deep OD,
is Algo and/or Hyperparameter Selection.

Need progress on Unsupervised Model Selection or Zero-shot Foundation Models.

Research fronts & Future:

In this talk:

I. Self-supervision

- Expressive aug. families
- Unsupervised val. losses
- End-to-end self-tuning paradigms

II. Foundation models/ in-context learning

- Complex data prior
- Scale-up for larger context
- Other data modalities

Not in this talk:

III. Meta-learning

- Bigger/better repositories
- Transfer beyond perf. predictors

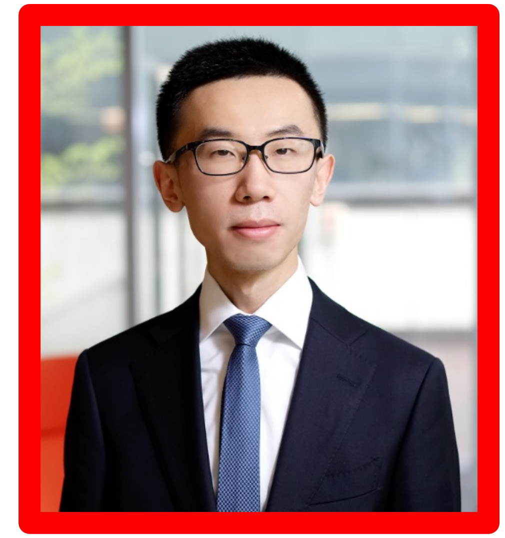
IV. Hyper-ensembles

- Speed & memory efficiency
- Selective ensembles

Students & postdocs (now prof.s):

Jaemin Yoo

Lingxiao Zhao



Yue Zhao

Yuchen Shen

Haomin Wen

Thank you!